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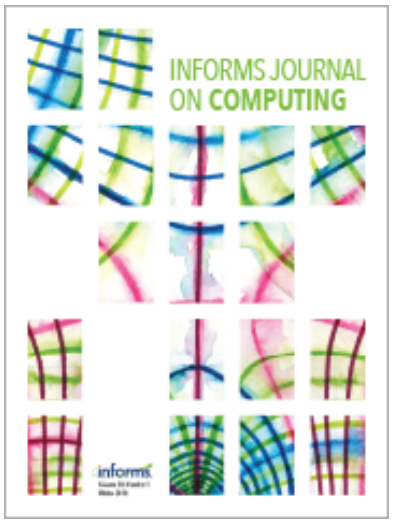

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Integrated Backup Rolling Stock Allocation and Timetable Rescheduling with Uncertain Time-Variant Passenger Demand Under Disruptive Events

Jiateng Yin,[a] Lixing Yang,[a,*] Andrea D'Ariano,[b] Tao Tang,[a] Ziyou Gao[a,*]

[a] State Key Laboratory of Rail Traffic Control and Safety, Beijing Jiaotong University, 100044 Beijing, China; [b] Department of Engineering, Università degli Studi Roma Tre, 00154 Rome, Italy
*Corresponding author

Contact: jtyin@bjtu.edu.cn (JY); lxyang@bjtu.edu.cn, https://orcid.org/0000-0003-1628-5015 (LY); andrea.dariano@uniroma3.it (AD); ttang@bjtu.edu.cn (TT); zygao@bjtu.edu.cn (ZG)

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Abstract. Railway traffic management focuses on regulating train movements and delivering improved service quality to passengers; however, such efforts are subject to many uncertainties in terms of disruptions and passenger demand on a rail transit line. In contrast to most existing studies, which focus on the rescheduling of passenger timetables in a deterministic framework, this study proposes a two-stage stochastic optimization model for allocating backup rolling stocks (BRS) to storage lines to reschedule the timetable and serve passengers delayed by disruptions. The first stage is an assignment problem to determine the optimal plan for the allocation of BRS to storage lines to achieve a good trade-off between the investment cost for the BRS and the expected travel time of delayed passengers across different stochastic scenarios. The second stage is explicitly formulated as a network flow model to optimize the timetable of the delayed trains on the tracks and the BRS from the storage lines such that the passenger travel time is minimized under each stochastic scenario. To improve the efficiency of convergence, we develop an improved L-shaped method with several accelerating techniques. Among these, we show that the classical integer L-shaped cut can be tightened given the property of the second-stage problem, which can also be generalized to other two-stage integer stochastic programs. Real-world case studies based on historical data from the Beijing metro verify the effectiveness of the proposed approach in reducing the travel time for passengers.

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Keywords: mathematical programming • backup rolling stock allocation • train rescheduling • two-stage stochastic optimization • Benders decomposition

1. Introduction

The urban rail system is a critical component of public transport in a large city because of its unique characteristics of low carbon emission, high carrying capacity, and punctuality. However, unexpected disruptions in daily operations often occur that may cause severe delays of multiple trains, partial blockages of lines, and passenger overcrowding on trains and at stations, which greatly reduce the service quality of rail systems (Cox et al. 2006).

For recovery from such unexpected events, the current strategy of rail dispatchers is to make train timetable rescheduling plans to alleviate the crowds of passengers waiting on platforms and recover the system to the normal state, for either metro or conventional railway lines (Louwerse and Huisman 2014, Lai and Leung 2018, Huang et al. 2020). As an alternative to such train rescheduling strategies, another effective and practical way to address a large number of delayed passengers in a crowded situation is to use backup rolling stocks (BRS) that are proactively allocated to storage lines (or side/pocket tracks at rail stations), as done, for example, in the Beijing metro (http://m.people.cn/n4/2020/0323/c902-13804959.html, accessed July 2022) and the Toronto subway (https://en.wikipedia.org/wiki/Line_1_Yonge-University, accessed July 2022). From their

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experience, rail dispatchers forecast the spatial-temporal distribution of passengers and the probability of service delays throughout the next day and allocate BRS to storage lines for potential use. During the operation of a rail system, BRS can help dispatchers reschedule the train timetable, quickly clear crowded platforms, and improve the service quality for passengers once a disruptive incident has been resolved and the required service has resumed (Yin et al. 2017b). In other words, the *resilience* of rail systems can be improved with off-line located BRS. This strategy has been particularly significant during the COVID-19 pandemic to control the passenger density on platforms (Rader et al. 2020). Nevertheless, because of the high cost of using BRS (rolling stock investment cost and payment of drivers' salaries), an outstanding challenge for rail managers is the lack of quantitative decision support for determining an optimal allocation plan for BRS and optimizing the rescheduled train timetable including BRS in the case of an unexpected event so as to balance the rolling stock investment cost and the service quality for passengers.

Although the integrated BRS allocation and timetable rescheduling problem (BARP) is a highly relevant practical problem in many rail transit systems in China and other countries, to the best of our knowledge, this problem has not been well addressed in the existing literature. The difficulty of this problem can be attributed to the following factors.

1. In theory, it is difficult to judge what characterizes a "good" BRS allocation plan, because of the widely varying nature of potential disruptive events in terms of their locations of occurrence, durations, and influences. Because BRS are used in response to disruptive events, it is critical to concurrently consider the allocation of BRS and the train rescheduling strategies to develop a high-quality and low-cost BRS allocation plan. Although a large number of studies have addressed the train rescheduling problem (e.g., the studies of Cacchiani et al. (2014), Lamorgese et al. (2018), Caimi et al. (2011), and Veelenturf et al. (2015)), there is no literature that directly addresses the BARP problem.

2. Compared with the passenger flows in conventional railway systems, those in congested urban rail transit systems have different characteristics because passengers do not usually consult the train timetables before their trips, leading to dynamic and uncertain passenger demand. In practice, it is difficult to formulate an efficient decision-making modeling approach that can simultaneously account for these dynamic and uncertain factors.

1.1. Contributions and Overview

(*The art of*) *managing is* (*in*) *foreseeing*. (Birge and Louveaux 2011, p. xi)

On the basis of a vast amount of historical data recorded for the Beijing metro, in this study, we aim to construct a stochastic programming model, to develop a decomposition-based algorithm, and to implement our methodology for real-world applications arising from a practical problem encountered in congested urban rail systems.

The three key contributions of this paper are summarized as follows:

- In contrast to the existing literature that addresses the train rescheduling problem on the operational level, to our knowledge, our study is the first to formalize the train rescheduling problem with BRS by combining the tactical and operational decisions necessary for the disruption management of urban rail systems in China. We propose the BARP to determine the optimal BRS allocation plan, which is evaluated by the BRS investment cost and the expected passenger travel time through considering the train rescheduling decisions in cases of disruptions, such as the blockages of certain tracks/trains or the overcrowding of passengers at stations.
- We extend the formulation proposed by Yin et al. (2018) to an integrated flow-based two-stage stochastic programming model for solving the BARP on the basis of a spatial-temporal network. Our model explicitly considers both the dynamic and uncertain characteristics of the passenger demands during train rescheduling for a rail transit system in cases of disruptive events. In addition, we introduce a new way to model the movements of trains in the network by introducing a set of *arc-occupation* variables instead of using an additional index for each train (Caprara et al. 2002, Niu and Zhou 2013). This new formulation can markedly reduce the number of decision variables while still capturing the specific constraints needed for modeling train movements in urban rail systems, such as headway constraints and train loading capacity constraints.
- By investigating the theoretical properties of the formulated BARP models, we develop efficient algorithms on the basis of Benders decomposition with a number of customized speed-up techniques motivated by recent literature contributions (e.g., in Angulo et al. (2016) and Rahmaniani et al. (2017)). A tightened version of classical integer L-shaped cut (see, e.g., Laporte and Louveaux 1993) is developed based on the model property that adding additional BRS improves the optimal value of recourse function, and the cut can be used in other similar problems. The proposed algorithms enable us to generate near-optimal solutions by adopting parallel computing techniques. Several sets of experimental results, based on practical railway line data and real-world passenger origin-destination data from the Beijing metro network, show that our methodology offers remarkable improvements compared with state-of-the-art solvers and practical train dispatching strategies. Moreover, our results have provided practical support

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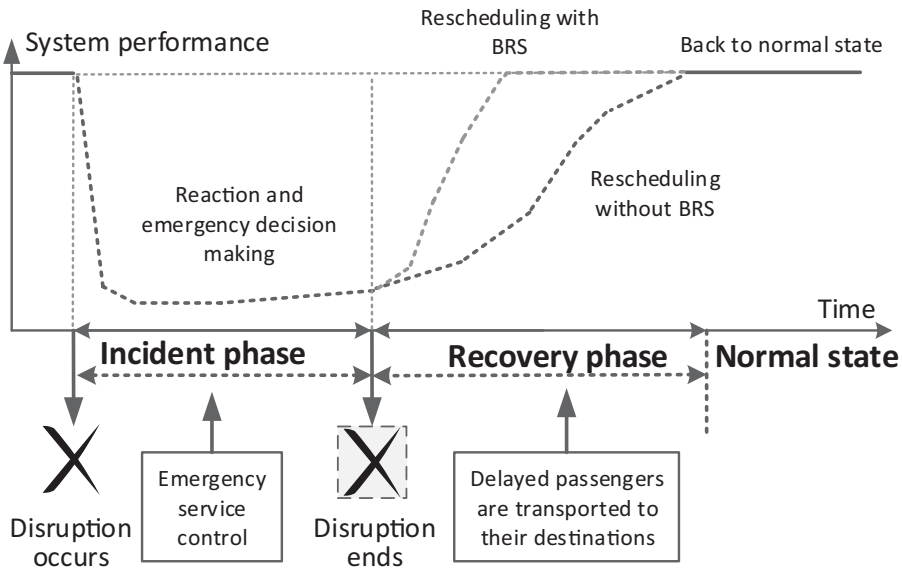
Figure 1. Phases of a Disruptive Event in an Urban Rail System



to the Beijing metro's decision makers, who began to allocate BRS to corresponding stations in May 2019.

The rest of this paper is organized as follows. Section 2 presents a detailed description of the studied problem. Section 3 reviews the previous works on closely related problems. Section 4 presents two BARP models and analyzes their theoretical properties. Section 5 introduces our improved integer L-shaped method for efficiently solving the models presented in Section 4. Section 6 reports several sets of computational experiments based on real-world data from the Beijing metro. Section 7 gives our conclusions and remarks on the potential use of the proposed methodology in practice.

2. Problem Description

In the context of urban rail systems, a disruptive event typically refers to the occurrence of an unexpected incident (e.g., a temporary track blockage) that makes it impossible to operate the schedule as it was originally planned. Typically, such a disruption consists of two main phases, the incident phase and the recovery phase, as displayed in Figure 1. In the incident phase, the rail manager must take actions quickly to avoid congestion of the trains, whereas in the recovery phase, the rail manager makes a new train schedule to smooth out the passenger volume, which has become oversaturated as a result of the disruption, as fast as possible. We focus on the latter phase to reduce the time required to recover the oversaturated passenger volume to the normal state (i.e., resilience against disruptive incidents) by means of BRS. In fact, for the Beijing metro, the development of tactical BRS allocation plans has been under consideration since 2014. With the preallocation of BRS in the network, rail managers can generate a rescheduled train timetable using the preallocated BRS to quickly clear crowded platforms during the recovery phase of a disruptive event.

In this study, we consider a typical urban rail line connecting a set of rail stations denoted by I. We focus on a tactical problem in which the rail managers plan to allocate a maximum fleet size K_b of BRS on a set of storage lines corresponding to a subset of stations denoted by I^S. As shown in Figure 2, the preallocated BRS are used when developing a rescheduled train timetable during the recovery phase of a disruptive

Figure 2. Illustration of the Train Rescheduling Process with the Potential Allocation of BRS

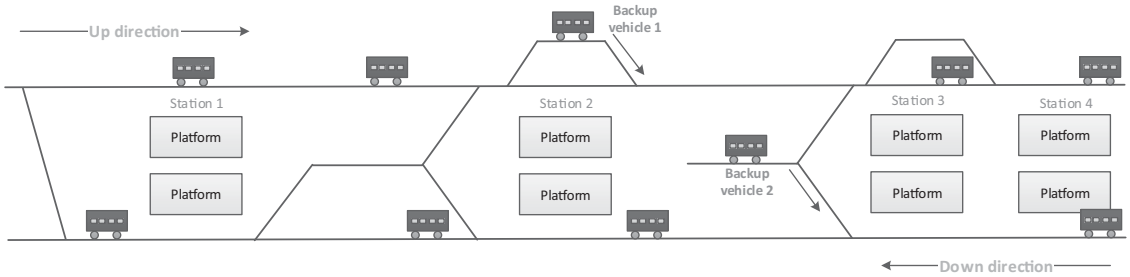


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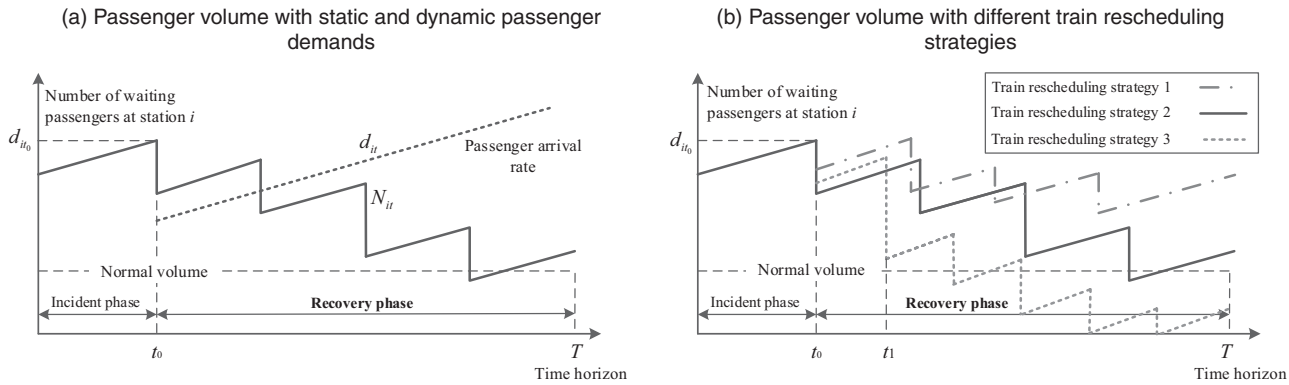
Figure 3. An Illustration of Passenger Volume Variations During the Recovery Phase



event to serve the delayed passengers and quickly clear crowded platforms. Nevertheless, the allocation of BRS requires an additional investment cost for the rolling stock and crew resources (e.g., the investment cost for purchasing one physical train vehicle for the Beijing metro is approximately ¥42 million). Thus, the objective of this study is to provide quantitative support to help decision makers understand the trade-off between the investment cost and the quality of service (QoS) for delayed passengers and allocate BRS where they are most needed.

In this context, the QoS performance indicator refers to the travel time of the delayed passengers. To quantitatively model the QoS for passengers, we need to explicitly know the passenger demand distribution as input. However, the passenger demand in practice is typically uncertain, thus increasing the difficulty of determining the optimal BRS allocation plan. To consider this uncertainty, we have obtained historical data sets from the Beijing metro, which reflect all unexpected incidents on each rail segment as well as historical passenger origin-destination (OD) travel data recorded by the automatic fare collection (AFC) system. We have used these historical data to construct a number $|W|$ of stochastic scenarios to evaluate the distributions of passengers under unexpected events. Accordingly, we propose a *two-stage stochastic programming model*, in which the first-stage model optimizes the tactical BRS allocation plan for a given number of stochastic scenarios, and the second-stage model is used to evaluate the first-stage decisions for each scenario with a deterministic passenger demand distribution (similar to an operational problem).

In the second-stage model, the uncertain factors are realized for each scenario. Then, with the deterministic passenger demand and the allocation plan for the BRS taken as input, the rescheduled train timetable is optimized to accurately *couple* the movements of the trains with the passenger demands to reduce the deterioration in QoS perceived by the passengers. For clarity, we divide the passenger demands in each scenario into two types: static and dynamic passenger demands. The static passenger demand refers to the number of passengers detained when the BRS are inserted into the rescheduled timetable, whereas the dynamic passenger demand reflects the passengers who continue to enter the stations during the recovery phase. Let T denote the length of the considered time horizon in the recovery phase, let d_{ijt} denote the number of passengers arriving at time t with origin station i and destination station j, and let N_{it} represent the number of passengers waiting at station i at time t. As shown in Figure 3(a), $d_{it_0} = \sum_{j\in I} d_{ijt_0}$ is the static passenger demand, which reflects the number of delayed passengers at station i at the initial time t_0, and $d_{it} = \sum_{j\in I} d_{ijt}$ is the dynamic passenger demand at station i at time t ($t \neq t_0$). Figure 3(b) illustrates the potential effects on passenger volume variations at station i under the following three train rescheduling strategies: in strategy 1, a nonoptimized train rescheduling plan with no BRS is used; in strategy 2, an optimized train rescheduling plan is implemented in accordance with the passenger distribution (Ortega et al. 2018); and in strategy 3, an integrated optimized train rescheduling plan with a preallocated BRS is adopted. We can see from Figure 3 that when a BRS vehicle is introduced into the rescheduled timetable at time t_1 (indicated by the dotted gray line), the waiting passenger volume is evidently decreased, because this vehicle is empty and can transport a large number of delayed passengers.

The QoS indicator is equivalent to the expected value of service quality for passengers across all stochastic scenarios. Using the uncertain and dynamic passenger demand data as input, BARP aims to determine the optimal BRS allocation plan (i.e., the fleet size and positions of the BRS) and the rescheduled train timetable to maximize the QoS with the minimal investment cost for allocating BRS.

2.1. Assumptions

To formulate the described problem as a rigorous mathematical model, we adopt the following assumptions throughout this paper.

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Assumption 1. *Each storage line is connected to one station and can be allocated at most one train at a given time to satisfy the requirements of the interlocking system.*

Assumption 2. *As a result of the station layout, neither overtaking nor meeting of trains occurs on metro lines, as trains usually run in first-in, first-out order from the first station to the final station.*

Assumption 3. *The QoS is evaluated in terms of the expected passenger travel time across stochastic scenarios. The travel time for each passenger is measured from his or her arrival time at the origin station to his or her alighting time at the destination station.*

Typically, Assumptions 1 and 2 are valid for urban rail transit lines because the station layout is relatively simple, as in the Beijing metro system (Yin et al. 2016). These two assumptions also allow us to use macroscopic models, in which each station is treated as a single capacitated node (in contrast to microscopic models, in which all nonshareable resources are represented, and the atomic movements of the trains are associated with each resource on their routes; see Lamorgese et al. (2016)) to formulate our studied problem. Regarding Assumption 3, we note that the travel time of the delayed passengers is a key factor in reducing the QoS (see, e.g., Wang (2014)), and this indicator is particularly important for reducing the congestion on platforms after disruptive events (especially during the COVID-19 pandemic, to avoid the risks of virus spread). Thus, this assumption is consistent with the requirements of rail managers.

3. Related Works

3.1. Literature on Train Timetable Rescheduling

In the field of railway disruption management, an extensive body of literature has been devoted to *responsive approaches*. These approaches usually consider an initial schedule, which is then adjusted in real time in response to disturbances and/or disruptions to recover the system to the normal state (see, e.g., Cordeau et al. (1998), Törnquist and Persson (2007), Cacchiani et al. (2014), Lamorgese et al. (2016), Lai and Leung (2018), Van der Hurk et al. (2018), and Lamorgese and Mannino (2019)). To reduce the negative effects caused by small disturbances or large disruptions on trains and passenger flows, numerous mathematical models and solution methodologies have been proposed for optimizing various objectives concerning, for example, the number of canceled trains (Zhan et al. 2015), delay time of trains (Corman et al. 2010, Lamorgese et al. 2016), passenger satisfaction (Sánchez-Martínez et al. 2016), energy consumption (Scheepmaker et al. 2017), or a trade-off among multiple objectives (Dollevoet et al. 2014b).

Mascis and Pacciarelli (2002) first investigated the job-shop scheduling problem with blocking and no-wait constraints, and this work was followed by a series of studies addressing specific train rescheduling problems (D'Ariano et al. 2007; Mannino and Mascis 2009; Liu and Kozan 2011; Lamorgese et al. 2016, 2018). For example, Mannino and Mascis (2009) developed a real-time train control model based on job-shop scheduling for regulating trains at metro stations. Their aim was to avoid potential conflicts between trains and to improve the punctuality and regularity of rail traffic. Using a branch-and-bound algorithm for train rescheduling embedded with a local search algorithm for local train rerouting, D'Ariano et al. (2008) and Corman et al. (2010) proposed a real-time traffic management system (ROMA), which was compared with simple train dispatching rules. Overall, most of these studies have addressed the difficulty of solving models on a real-world scale within strict computational time limits as a result of the large number of 0-1 variables required to manage the train timing, ordering, and routing decisions. Among these solution algorithms, a very popular approach is the macroscopic/microscopic decomposition approach (Lamorgese et al. 2018), in which the station capacity is initially ignored to generate a train schedule using a macroscopic model, and the feasibility of the schedule is then evaluated by considering the microscopic station track layout. Lamorgese and Mannino (2015) and Lamorgese et al. (2016) first proposed an exact algorithm inspired by Benders decomposition, which was compared with a first-in, first-out (FIFO) heuristic and has been applied to a Norwegian railway network since 2014. The key idea is to decompose the problem into a line dispatching problem (*restricted master problem*) and several station rerouting subproblems. This treatment allows the model to be efficiently solved to the (near-)optimal solution by iteratively solving the master problem and the subproblems and then adding a set of Benders cuts into the master problem.

Two main distinctions can be drawn between urban rail transit lines and conventional railway systems. First, the railway track layout of transit lines is much simpler than that of conventional railways, and train reordering/rerouting decisions are usually not allowed on transit lines; second, the dynamic characteristics of the passenger flows in congested urban rail transit systems are much more evident than those in conventional railway systems (Yin et al. 2021). Because of these two features, macroscopic approaches considering time-varying passenger flows are preferred in the current literature for addressing train scheduling and rescheduling problems for urban metro lines. For timetable design with time-varying passenger demands in a metro corridor, Barrena et al. (2014) developed nonlinear mathematical formulations to minimize the passenger wait time at stations, and a fast adaptive large neighborhood search algorithm was proposed to obtain a good-quality solution. Yin et al. (2016) proposed an efficient approximate dynamic

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programming approach for real-time metro train rescheduling. The authors considered the metro system to be affected by uncertainty in passenger demand, which was modeled as a time-dependent Poisson distribution for each station on the line. Ortega et al. (2018) addressed the metro train rescheduling problem in the case of a fleet size reduction as a result of a disruption, and the impact on the passengers was modeled via a passenger inconvenience function. Yin et al. (2017b, 2018) introduced a practical metro train rescheduling strategy with the BRS and unlimited train boarding capacity. To solve the corresponding problem, they proposed a real-time heuristically iterative algorithm. A small-scale case study was conducted on the Yizhuang line of the Beijing metro. Lai and Leung (2018) developed several mathematical models for adjusting schedules in real time in accordance with train feedback information. A rolling horizon approach embedded with an event-based simulation model was proposed by the authors for improving computational efficiency.

3.2. Literature on Preventive Planning for Disruptive Events

Whereas most existing studies have focused on responsive approaches for dealing with disruptive events, only a limited amount of research has been published on tactical planning for preventing disruptions and improving the ability to recover from disruptions as quickly as possible. However, these topics are now receiving increasing attention (Alderson et al. 2014, Ganin et al. 2017). Recently, *preventive planning* strategies and models have been proposed for designing transport services that are robust and resilient to disturbances and/or disruptions (see, e.g., Cacchiani et al. (2012a, 2012b), Pender et al. (2012), Jin et al. (2016), and Wang and Jacquillat (2020)).

To improve the robustness (defined as the ability to recover from disruptions) of rolling stock circulation for Netherlands Railways, Cacchiani et al. (2012b) developed a two-stage stochastic programming approach for optimizing the tactical rolling stock circulation plan as well as a recovery solution (i.e., a rolling stock rescheduling decision) for each disruption scenario. Approximately 30 disruption scenarios were generated as the basis for developing a robust rolling stock circulation and rescheduling plan. A Benders decomposition–based heuristic was designed to solve their stochastic programming model. To improve the capability of a transport network to recover from potential disruptions, Chen and Miller-Hooks (2012) proposed a stochastic mixed-integer programming formulation. Their model seeks an optimal set of recovery activities to be undertaken in the immediate aftermath of a disruption such that the expected number of shipments can be maximized within a given budget. Wang and Jacquillat (2020) proposed a two-stage stochastic programming approach for the tactical air traffic management under uncertainty arising from weather and other unexpected conditions. A set of dual integer cuts incorporated with neighborhood constraints were proposed to improve the computational efficiency. Considering the uncertainty of passenger demand, Cacchiani et al. (2020) proposed the concept of *light robustness* to make a tactical train schedule with optimized stopping patterns. Their focus is on a robust solution in the planning phase, so as to reduce the passenger inconvenience that may occur in real time as a result of additional passenger demand. A similar idea was followed in Pu and Zhan (2021), which addressed the railway line planning problem given uncertain passenger demand.

In regard to urban transit systems, some recent studies have highlighted the importance of optimizing bus service networks to withstand and quickly recover from unexpected disruptive events in metro systems. For example, with the aim of reducing the negative effects of potential disruptive events in the Singapore metro network, Jin et al. (2014) proposed a design for a new bus network to enhance the resilience of the intermodal bus-metro transit network. They developed a two-stage stochastic programming model in which the aim of the first stage is to optimize the bus network design, and in the second stage, the optimal passenger flows are generated under each disruption scenario. However, in that study, it was assumed that a metro line cannot transport any passengers at all in the case of a disruption; hence, an in-depth analysis of recovery strategies for the train traffic flows (i.e., the train rescheduling strategies required during the recovery phase) was lacking.

Table 1 presents a comparison of our study and other closely related literature. Similar to the idea adopted in Cacchiani et al. (2012b) and Wang and Jacquillat (2020), a two-stage stochastic programming approach is employed in our study to address an integrated tactical planning problem arising in practical applications. Different from Cacchiani et al. (2012b), who address the mainline railway rolling stock circulation problem (without consideration of passenger flows), to our knowledge, we are the first to integrate tactical BRS allocation and operational timetable rescheduling in urban rail transit systems into comprehensive stochastic optimization models that consider uncertain time-dependent passenger demands. We also develop an improved L-shaped method based on Benders decomposition, which includes a number of customized speed-up techniques based on an analysis of the mathematical properties of our formulations.

4. Mathematical Formulations

In this section, we formally introduce the construction of a spatial-temporal network for the considered rail transport system. Then, we develop two mixed-integer

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Table 1. Detailed Comparison of Studies That Are Closely Related to Our Work

Publication	System (railway vs. transit)	Decisions[a]	Passenger characteristics[b]	Solution method
Lamorgese and Mannino (2015)	Railway	Rescheduling and rerouting (O)	—	Benders decomposition
D'Ariano et al. (2007)	Railway	Timetable rescheduling (O)	—	Branch-and-bound
Dollevoet et al. (2014a)	Railway	Timetable rescheduling and passenger rerouting (O)	Time-dependent OD pairs	Iterative heuristic
Wong et al. (2008)	Transit network	Train timetabling (T)	Piecewise constant (transfer)[c]	Optimization-based heuristic
Lai and Leung (2018)	Transit line	Train rescheduling (O)	Dynamic[d]	Rolling horizon optimization
Huang et al. (2020)	Transit line	Timetable rescheduling (O)	Dynamic	Two-stage optimization
Cacchiani et al. (2012b)	Railway	Rolling stock plan with adjusted strategy (T+O)	Static	Benders decomposition–based heuristic
Jin et al. (2014)	Transit network	Bus network design (T)	Uncertain and static	CPLEX
Yin et al. (2018)	Transit line	Timetable rescheduling with BRS (O)	Static	Heuristic
This paper	Transit line	BRS allocation with timetable rescheduling and passenger flows (T+O)	Uncertain and dynamic	Improved L-shaped method

[a]Symbolic descriptions of decisions: operational decisions (O), tactical planning (T), and tactical planning evaluated with operational decisions (T+O).
[b]The em dash (—) indicates that passenger characteristics are not considered in the corresponding study.
[c]This study addresses train timetabling in a rail transit network, considering passenger transfers.
[d]In this study, the demands of the passengers are captured by the coverage of train frequency.

linear programming (MILP) formulations for the BARP. Finally, we analyze the mathematical properties of the constructed models. All of the symbols, notation, and variables used are summarized in Table A.1 in the appendix.

4.1. Construction of the Spatial-Temporal Network

To model the movements of trains and the dynamic passenger flows in the recovery phase of a disruptive event, we first discretize the considered time horizon into a set of discrete timestamps $\{0, 1, \ldots, T\}$, denoted by $\mathbb{T}$ (where timestamp 0 represents the initial time t_0 of the recovery phase in Figure 3). Then, we construct a spatial-temporal network $G(N, A)$, in which each node $n_{it} \in N$ represents a station $i \in I$ at timestamp $t \in \mathbb{T}$ and each arc in A represents a spatial-temporal movement of a train (or a number of passengers) from one station to another between an earlier period and a later one. Three types of arcs are defined in set A, as follows:

- A waiting arc $(n_{it}, n_{it'}) \in A^W$ is defined for each pair $n_{it}, n_{it'} \in N$, where $t' = t + 1$ (Yin et al. 2017a). A flow on one of these arcs represents a train (or a number of passengers) dwelling at station i from timestamp t to timestamp t'. The introduction of waiting arcs is critical for formulating the movements of trains and passengers in oversaturated situations, in which passengers must wait for more than one train.
- A travel arc $(n_{it}, n_{js}) \in A^T$ is defined for each pair $n_{it}, n_{js} \in N$, where $j = i + 1$, $s = t + t_{ij}$, and t_{ij} denotes the travel time of a train between stations i and j, because we assume that the trains are not allowed to skip any station or change their routes. A flow on one of these arcs represents a train (or a number of passengers) traveling from station i at time t to station j at time s.
- An assignment arc $(n_O, n_{it}) \in A^I$ is defined for each $n_{it} \in N$ $(i \in I^S)$, where n_O is a dummy origin in the constructed network. A flow on one of these arcs represents the assignment of a BRS vehicle to the corresponding spatial-temporal node n_{it}. Specifically, only stations with storage lines have assignment arcs, indicating that the allocated BRS at these stations can be added into the rescheduled timetable.

The set A is the union of the three types of arcs described in the preceding (i.e., $A = A^W \cup A^T \cup A^I$). Let $\delta^+(n_{it}) \subset A$ and $\delta^-(n_{it}) \subset A$ denote the sets of arcs for which n_{it} is the origin and destination node, respectively. Similarly, let $\sigma^+(n_{it}) \subset A^T$ and $\sigma^-(n_{it}) \subset A^T$ denote the sets of travel arcs for which n_{it} is the origin and destination node, respectively. We now illustrate the construction of a spatial-temporal network in the example presented in Figure 4, which shows the corresponding network over timestamps $t = 0, \ldots, 5$ and considers a rail transit line with two stations, i and j. The light gray lines represent the spatial-temporal trajectory of a BRS vehicle, which consists of two arcs: an assignment arc (n_O, n_{j2}) and a travel arc (n_{j2}, n_{i4}). From these two arcs, we can see that the vehicle is dispatched to station j at timestamp 2 and then travels to station i at timestamp 4.

We next discuss the two types of flows in the spatial-temporal network that are considered when

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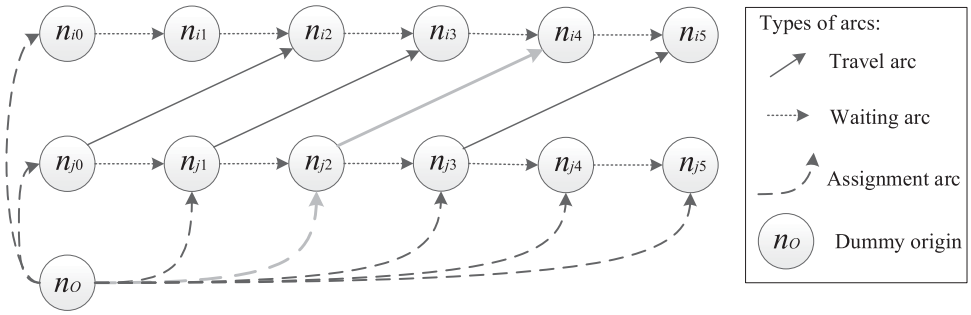
Figure 4. Example of a Spatial-Temporal Network



formulating the mathematical models. The first type models the flows of trains and is used to describe the movements of trains in the spatial-temporal network, represented by $\mathbf{x}$ in the following. In other words, the set of variables $\mathbf{x}$ defines the rescheduled train timetable in the spatial-temporal formulation. The second type models the flows of passengers, denoted by $\mathbf{y}$ in the following. The latter set of variables describes the number of passengers who are traveling (or waiting) on each spatial-temporal arc. With this approach, we can integrate BRS allocation, timetable rescheduling, and passenger flow control in the recovery phase of a disruptive event (as shown in Figure 5).

4.2. A Two-Stage Stochastic Integer Programming Formulation

In our problem, the key decisions include the total fleet size of BRS that is employed and the number of vehicles allocated to the storage line connected to each station i, denoted by z_i, for all $i \in I^S$. According to Assumption 1, each storage line can be allocated at most one vehicle at a given time. The BRS allocation variables are thus defined as follows:

$$z_i = \begin{cases} 1, & \text{if a vehicle is allocated to storage line } i \\ 0, & \text{otherwise,} \end{cases} \quad \forall i \in I^S. \tag{1}$$

To capture the uncertain factors in our problem, we employ a two-stage stochastic programming approach, in which the binary vector $\mathbf{z}$ represents the set of first-stage decisions. Let c_i be the investment cost for allocating a BRS vehicle to storage line i. Then, we formulate the BARP as follows:

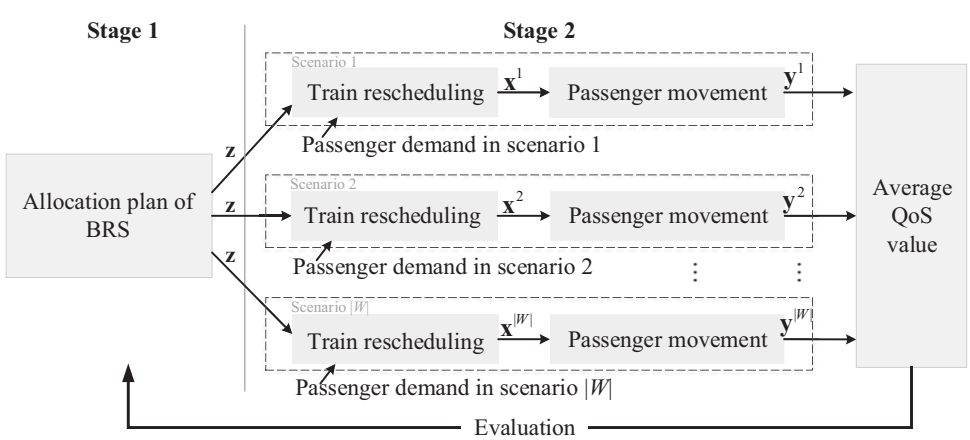
$$\min_{\mathbf{z}} \sum_{i \in I^S} \lambda_d c_i z_i + Q(\mathbf{z}) \tag{2}$$

$$\text{s.t.} \sum_{i \in I^S} z_i \le K_b, \tag{3}$$

$$z_i \in \{0, 1\}, i \in I^S. \tag{4}$$

This is essentially an assignment problem in which a set of BRS vehicles is allocated into the network to transport passengers or commodities. The constraints expressed in (3) ensure that the total fleet size of BRS initially allocated to the stations does not exceed the maximum number K_b (i.e., the maximum budget). The first term in the objective function in (2) represents the total investment cost for the allocation of BRS, whereas the second term refers to the expected QoS indicator under the uncertain demand given the first-stage decisions $\mathbf{z}$. Here, we define λ_d as a weight coefficient that reflects the importance of the investment cost. In this way, we transform the investment cost and the QoS into generalized costs expressed on the same unit scale

Figure 5. Framework of the Two-Stage Stochastic Programming Formulation Presented in This Paper



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in the objective function. In summary, we minimize two objectives: the total cost of allocating BRS and the expected QoS performance cost of the rescheduled timetable (see Section 6.3, which investigates the trade-off between these biobjectives).

To model the uncertainty of the passenger demands, this paper considers a given number of demand scenarios drawn from historical passenger data from the AFC center of the Beijing metro system, which stores all passengers' travel data for the studied transit line. We index these discrete scenarios by $w \in W$; the passenger OD data in scenario w are denoted by $\mathbf{d}(w) = [d^w_{ijt}, i,j \in I, t \in \mathbb{T}]$, and the probability of occurrence of scenario w is denoted by p^w, $w \in W$. In this way, the second-stage model is formulated as a network flow model to address a passenger-oriented train rescheduling problem given the allocated BRS, as described in the following subsections.

4.2.1. Parameters and Variables in the Second Stage.

First, for clarity of the formulation, we focus only on the static passenger demand (i.e., at timestamp $t = 0$) in the second-stage model. In other words, this subsection does not address the dynamic passenger demand and assumes that no new passengers are allowed to enter stations during the recovery phase (i.e., at $t \in \{1,2,\ldots,T\}$). In this situation, the second-stage model is used to optimize the passenger and train flows in the spatial-temporal network given the BRS allocated to the storage lines $i \in I^S$ in accordance with random passenger demands. Because we consider only the static demand, we can use d^w_{ij} (substituted for d^w_{ijt}, where $t = 0$) for $i,j \in I$ to represent the number of passengers in each scenario $w \in W$. Then, we define parameters $B^w_i = \sum_{j\in I} d^w_{ij}$ and $L^w_j = \sum_{i\in I} d^w_{ij}$ to represent the passenger demands with origin i and destination j, respectively. In addition, we define two sets of timestamps for each $t \in \mathbb{T}$—that is, $\mathbb{T}_t = \{0,1,\ldots,t\}$ and $\mathbb{T}^{\min}_t = \{\tau \in \mathbb{T} | t \le \tau \le t + t_{\min}\}$, where $t_{\min}$ represents the minimum headway time in our mathematical formulation.

The purpose of our second-stage model is to generate an optimally rescheduled timetable with a given fleet of BRS such that the QoS (i.e., the travel time of the affected passengers in accordance with Assumption 3) is minimized. We define three sets of decision variables to formulate this problem under each discrete scenario w. The first set of variables, $x^w_a \in \{0,1\}$, the arc-occupation variables for each arc $a \in A$, describes the movements of the trains in the network (here, notation a represents the spatial-temporal arc for simplicity, and we will henceforth use arc-based notations for modeling convenience). Each of these variables models whether the corresponding arc a is occupied by a train in the rescheduled timetable. The second set of variables is the set of passenger flow variables, $y^w_a \in \mathbb{R}$. Each variable y^w_a represents the number of passengers traveling on the corresponding arc a, where $a \in A^W \cup A^T$ in the spatial-temporal network. The third set of variables is denoted by α^w_{it} for $i \in I$, $w \in W$, and $t \in \mathbb{T}\backslash\{0\}$. Each variable α^w_{it} represents the numbers of passengers alighting at the corresponding spatial-temporal node $n_{it} \in N$. Here, in contrast to most recent mathematical models for train scheduling and rescheduling (e.g., Caprara et al. (2002), Niu and Zhou (2013), and Cacchiani et al. (2014)), which use an additional index k to represent each train, a novel mathematical formulation is proposed that does not consider the detailed spatial-temporal train/passenger trajectories but rather considers only the flows of trains (i.e., $\mathbf{x}$ consisting of variables x^w_a, $a \in A$, $w \in W$) and passengers (i.e., $\mathbf{y}$ consisting of variables y^w_a, $a \in A$, $w \in W$) in the spatial-temporal network. In this way, our methodology offers the great advantage of markedly reducing the number of decision variables, especially for instances with high traffic density (in which a large number of trains must be considered). The limitation is that this kind of formulation faces difficulties when modeling train overtaking and crossing because of several big M variables are required to determine the order of the trains. However, because we focus on rail transit lines, where train overtaking and crossing operations are not allowed (see Assumption 2), our mathematical formulation is ideally suited for the studied problem.

4.2.2. Constraints and Objectives in the Second Stage.

After defining the decision variables, we next introduce the following four sets of constraints for each scenario $w \in W$ to guarantee that a set of train movement variables x^w_a ($a \in A$) determines a feasible train schedule in which neither overtaking nor crossing is allowed, and that the passenger flows in the network are exactly matched with the movements of the trains.

Constraint set 1: Train flow balance constraints:

$$\sum_{a\in\delta^+(n_{it})} x^w_a - \sum_{a\in\delta^-(n_{it})} x^w_a = \begin{cases} 1 & \text{if} \quad t = 0, i \in I^E \\ 0 & \text{otherwise,} \end{cases} \quad \forall t \in \mathbb{T}\backslash\{T\}, i \in I, w \in W. \tag{5}$$

Constraint set 2: Assignment constraints for BRS:

$$\sum_{t\in\mathbb{T}} x^w_a \le z_i, \quad \forall i \in I^S,\ a = (n_O, n_{it}) \in A^I, w \in W. \tag{6}$$

Constraint set 3: Headway constraints to guarantee the safety of train operations:

$$\sum_{\tau\in\mathbb{T}^{\min}_t} \sum_{a\in\sigma^-(n_{i\tau})} x^w_a \le 1, \quad \forall i \in I,\ t \in \mathbb{T}, w \in W. \tag{7}$$

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Constraint set 4: Passenger assignment and train capacity constraints

(a) Passenger flow constraints

$$\sum_{a\in\delta^-(n_{it})} y_a^w - \sum_{a\in\delta^+(n_{it})} y_a^w = \begin{cases} -B_i^w & \text{if } t=0 \\ \alpha_{it}^w & \text{otherwise,} \end{cases} \quad \forall i\in I, t\in\mathbb{T}, w\in W. \tag{8}$$

(b) Passenger alighting constraints

$$\sum_{t\in\mathbb{T}\setminus\{0\}} \alpha_{it}^w \le L_i^w, \qquad \forall i\in I, w\in W. \tag{9}$$

$$\alpha_{it}^w \le \sum_{a\in\sigma^-(n_{it})} x_a^w \cdot M, \qquad \forall i\in I, t\in\mathbb{T}\setminus\{0\}, w\in W.$$

通过alighting的乘客数连接变量x和y

$$\tag{10}$$

(c) Train capacity constraints

$$0 \le y_a^w \le C, \qquad \forall a\in A^T, w\in W. \tag{11}$$

The first three groups of constraints guarantee the feasibility of the rescheduled train timetable in the spatial-temporal network given the BRS allocation plan from the first stage. Specifically, Equation (5) gives the flow balance constraints regarding the variables $\mathbf{x}$ that enable the commodity (i.e., train) flows in the spatial-temporal network. At timestamp $t = 0$, the *existing* trains (physical vehicle trains that are already in the network at the beginning of the recovery phase) are assigned to a set of stations I^E, whereas at timestamps $t\in\{1,\ldots,T-1\}$, the trains (involving BRS) use waiting, travel, and assignment arcs to move toward their destinations. Note that the flow balance constraints are defined on the set $\{0,1,\ldots,T-1\}$, reflecting that the nodes n_{iT} for $i\in I$ actually represent a set of dummy destinations for the trains, and there is only inbound train flow into these nodes. Equation (6) presents the assignment constraints utilized to insert the BRS into the rescheduled timetable. Equation (7) specifies the headway capacity constraints applied to guarantee that each time window $\mathbb{T}_t^{\min}$ can accommodate at most one train, which is done by ensuring that the trains are separated by at least a minimum headway time $t_{\min}$.

Equations (8)–(11) are the constraints related to the passenger flows. Among these constraints, Equation (8) gives the commodity flow constraints for passengers. Constraints (9) and (10) guarantee that the passenger flows are subject to the OD data and train movements. Only when a train arrives can passengers alight; otherwise, α_{it}^w is equal to 0. A big M variable is introduced here to indicate whether a train arrives at spatial-temporal node $n_{it}\in N$. Equation (11) specifies the arc capacity constraints, which ensure that the number of passengers on each travel arc a is no more than the corresponding train capacity.

Regarding the objective function, we minimize the expected value of QoS considering the delayed passengers across different stochastic scenarios using a sample average approximation technique (Birge and Louveaux 2011). Because α_{it}^w refers to the number of passengers alighting at node n_{it}, we find that the number of passengers who have not reached their destinations by the last timestamp T is $\sum_{i\in I}(L_i^w - \sum_{t\in\mathbb{T}}\alpha_{it}^w)$. The average travel time for these passengers is set as T (i.e., the end of the considered time horizon). In addition, we have $\alpha_{it}^w = 0$ at time $t = 0$ because of Constraints (10). Thus, given a static passenger demand as input, we can formulate the objective function of the second-stage model as follows:

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$$Q(\mathbf{z}) = \min \sum_{w\in W}\sum_{i\in I} p^w\left(L_i^w T + \sum_{t\in\mathbb{T}\setminus\{0\}}(t\alpha_{it}^w - T\alpha_{it}^w)\right), \tag{12}$$

where the passenger traveling time under each scenario w is composed of two parts: $\sum_{i\in I}\sum_{t\in\mathbb{T}\setminus\{0\}} t\alpha_{it}^w$ represents the total travel time of passengers who have reached their destinations, and the other elements represent the travel time T of the remaining passengers.

4.2.3. MILP Formulation of BARP. The model described in Section 4.2.2 introduces the set of train rescheduling variables $\mathbf{x}$ and the set of passenger flow variables $\mathbf{y}$ that jointly determine the optimal numbers of boarding and alighting passenger at each spatial-temporal node $n_{it}\in N$. The idea that combines two kinds of flows is similar to Martin-Iradi and Ropke (2022) but has a different scope. In our formulation, α_{it}^w are actually a set of intermediate variables associated with $\mathbf{y}$ that essentially represent the passenger flow control strategy with minimized passenger travel time when the timetable is rescheduled.

In the objective function (12), because L_i^w represents the input data for the BARP, we know that minimizing (12) is equivalent to minimizing $\sum_{w\in W}\sum_{i\in I}\sum_{t\in\mathbb{T}\setminus\{0\}} p^w (t-T)\alpha_{it}^w$. In accordance with the equality constraints defined in (8), we can construct the following MILP formulation for the BARP with static demand (BARP-S):

$$\min_{\mathbf{x},\mathbf{y},\mathbf{z}} \sum_{i\in I^S}\lambda_d c_i z_i + \sum_{w\in W}\sum_{i\in I}\sum_{t\in\mathbb{T}\setminus\{0\}} p^w(t-T)\left(\sum_{a\in\delta^-(n_{it})} y_a^w - \sum_{a\in\delta^+(n_{it})} y_a^w\right) \tag{13}$$

$$\text{s.t.} \quad \sum_{a\in\delta^-(n_{it})} y_a^w - \sum_{a\in\delta^+(n_{it})} y_a^w : \begin{cases} = -B_i^w & \text{if } t=0 \\ \le \sum_{a\in\sigma^-(n_{it})} x_a^w\cdot M & \text{otherwise,} \end{cases} \quad \forall i\in I, t\in\mathbb{T}, w\in W, \tag{14}$$

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$$\sum_{t\in\mathbb{T}\backslash\{0\}}\left(\sum_{a\in\delta^-(n_{it})} y_a^w - \sum_{a\in\delta^+(n_{it})} y_a^w\right) \le L_i^w, \quad \forall i\in I, w\in W$$

and Constraints (4) – (7) and (11). (15)

4.3. Extensions of the Proposed Model

We now extend the above-mentioned BARP-S model to a more general formulation, in which the dynamic characteristics of passenger arrivals are taken into account. In the BARP-S model, which does not consider time-dependent passenger demands after a disruptive incident, we have the following equation:

$$\sum_{a\in\delta^-(n_{it})} y_a^w - \sum_{a\in\delta^+(n_{it})} y_a^w = \alpha_{it}^w, \quad \forall i\in I, t\in\mathbb{T}\backslash\{0\}, w\in W, \tag{16}$$

which represents how the delayed passengers are gradually served by the arriving trains in the spatial-temporal network. However, once we consider that there will also be new passengers entering the stations at every timestamp $t\in\mathbb{T}\backslash\{0\}$ during the recovery phase, the aforementioned passenger flow balance constraints should be revised to the following:

$$\sum_{a\in\delta^-(n_{it})} y_a^w - \sum_{a\in\delta^+(n_{it})} y_a^w = \alpha_{it}^w - d_{it}^w, \quad \forall i\in I, t\in\mathbb{T}\backslash\{0\}, w\in W, \tag{17}$$

where $d_{it}^w = \sum_{j\in I} d_{ijt}^w$ represents the number of passengers arriving at station i at timestamp t over scenario w.

Note that in this new formulation, we also need to restrict the number of alighting passengers at each station to ensure that only passengers who have boarded a train can reach their destinations. To this end, we extend the parameter L_i^w (i.e., the number of passengers alighting at station i in scenario w) to the new notation L_{it}^w, which is called the time-dependent passenger alighting parameter. The parameter L_{it}^w is essentially a cumulative plot that represents the number of passengers with destination i who plan to alight up to time t in scenario w. This parameter can also be computed in the form of an OD matrix generated via historical passenger demand data. Then, the constraints defined in (15) should be replaced by the following inequalities:

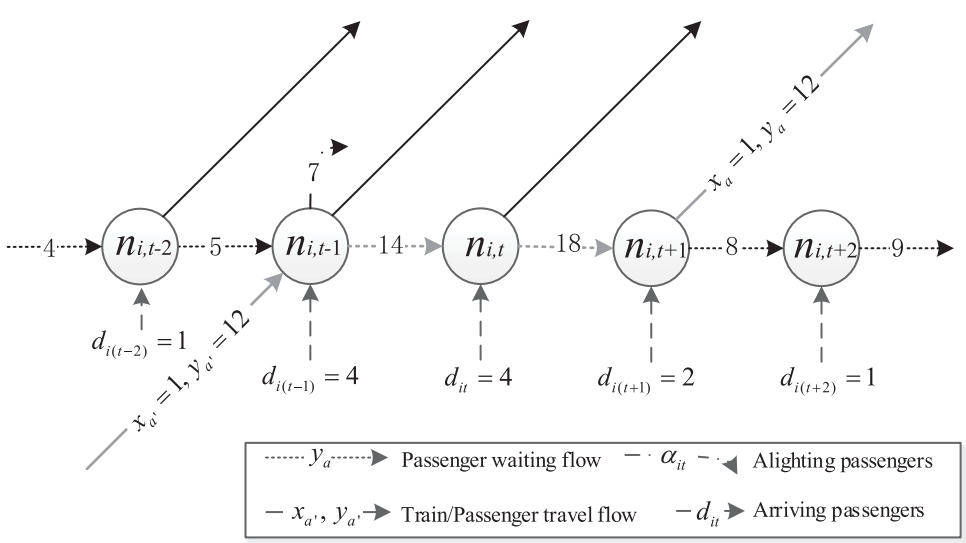
$$\sum_{\tau\in\mathbb{T}_t\backslash\{0\}} \alpha_{i\tau}^w \le L_{it}^w, \quad \forall i\in I, t\in\mathbb{T}\backslash\{0\}, w\in W, \tag{18}$$

where $\alpha_{i\tau}^w$ can be derived from Equation (17).

Example 1. Example 1, shown in Figure 6, illustrates a spatial-temporal network with five nodes, $n_{i,t-2},\ldots,n_{i,t+2}$, and it shows the passenger flow variations during the train arrival and departure processes. In this illustrative example, a train with capacity $C=12$ arrives at station i at timestamp $t-1$, dwells for two time units because of passenger alighting and boarding, and departs from station i at timestamp $t+1$, as shown by gray lines. Likewise, the black dashed lines represent the newly arriving passengers at each timestamp, whereas the dotted lines represent the flows of waiting passengers at station i. We can see from this example that when a train arrives at station i, a total of seven passengers can alight from this train at timestamp $t-1$.

We will next show how to formulate the passenger flow balance constraint at node $n_{i,t-1}$. According to Equation (17), we have $(12+5)-14=7-4$. We can see that the inflow volume of node $n_{i,t-1}$ is composed of 12 passengers arriving at this station and 5 waiting passengers. At this spatial-temporal node, 7 passengers leave the train, and the remaining capacity after they alight is 5. Then, the outflow of node $n_{i,t-1}$ is 14,

Figure 6. Example of Passenger Volume Variations with Dynamic Demands



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composed of 5 in-vehicle passengers and 9 waiting passengers. Because the remaining capacity of the train is 7, some waiting passengers can board the train before its departure from this station. This simple example illustrates how Equations (10), (17), and (18) work together to capture the correlations among the variable sets **x** and **y** and the dynamic characteristics of passenger arrivals in the spatial-temporal network. □

In summary, when considering time-dependent passenger arrivals on the basis of the BARP-S model, the two-stage stochastic programming model for BRS allocation and rescheduling with uncertain and dynamic passenger demand (BARP-D) can be formulated as follows:

$$\min_{\mathbf{x},\mathbf{y},\mathbf{z}} \sum_{i\in I^S} \lambda_d c_i z_i + \sum_{w\in W}\sum_{i\in I}\sum_{t\in\mathbb{T}\setminus\{0\}} p^w(t-T) \left(\sum_{a\in\delta^-(n_{it})} y_a^w - \sum_{a\in\delta^+(n_{it})} y_a^w\right) \tag{19}$$

s.t. Constraints (3)–(7), (11), (14), (17), and (18).

Because d_{it}^w represents the input data for the BARP-D, we know that minimizing (19) is equivalent to minimizing $\sum_{w\in W}\sum_{i\in I}\sum_{t\in\mathbb{T}\setminus\{0\}} p^w(t-T)\alpha_{it}^w$, the same as that of BARP-S.

4.4. Theoretical Properties of the BARP Models

The developed BARP-S and BARP-D models are both MILP formulations, with three types of variables (i.e., binary variables **z** and **x**, as well as nonnegative continuous decision variables **y**), linear equality (or inequality) constraints, and a linear objective function. We next investigate some of the theoretical properties of these BARP models, which will serve as the cornerstone for the development of efficient algorithms in the following section.

Lemma 4.1. *For a set of first-stage decision variables z_i, $i \in I^S$, that satisfy Equation (3), there always exists a set of feasible second-stage values* **x** *and* **y** *in the BARP-S and BARP-D models.*

Proof. Because the BARP-S model is a relaxed formulation of the BARP-D model (as described in Section 4.3), we present only the proof for BARP-D. Given a set of z_i with $i \in I^S$, we have $0 \le \sum_{t\in\mathbb{T}} x_a^w \le z_i$ for $i \in I^S$ and $a = (n_O, n_{it}) \in A^I$, according to Equation (6). In other words, it is a relaxed formulation of the special case that $\sum_{t\in\mathbb{T}} x_a^w = 0$, $\forall i \in I^S$, $a = (n_O, n_{it}) \in A^I$ (i.e., the case that no BRS are used). Under this special case, we consider a set of values **x**, where $\sum_{a\in\delta^+(n_{it})} x_a^w - \sum_{a\in\delta^-(n_{it})} x_a^w = 1$ for $i \in I^E$ and $t = 0$, and $x_a^w = 0$ for $a \in A^T$. This is a solution with waiting arcs only, in which the trains use waiting arcs to dwell at their corresponding stations. In this case, for each $i \in I$, we have $\sum_{a\in\delta^-(n_{it})} y_a^w - \sum_{a\in\delta^+(n_{it})} y_a^w = -B_i^w$ for $t = 0$, and $\alpha_{it}^w = 0$ and $\sum_{a\in\delta^-(n_{it})} y_a^w - \sum_{a\in\delta^+(n_{it})} y_a^w = -d_{it}^w$ for $t \in \mathbb{T}\setminus\{0\}$. In addition, this set of values **y** obviously satisfies Equations (17) and (18). Because this tighter formulation of the BARP-D model has feasible values **x** and **y**, we conclude that, for a set of first-stage decision variables z_i, $i \in I^S$, that satisfy Equation (3), the BARP-D model always has a feasible solution. □

We define $|\mathbf{z}|$ as the number of values in **z** that are equal to 1, and $f(|\mathbf{z}|) = \{\min_{\mathbf{z}} Q(\mathbf{z}) | \sum_{i\in I^S} z_i = |\mathbf{z}|\}$. Here, $f(|\mathbf{z}|)$ essentially refers to the optimal cost of the second-stage BARP model given a total of $|\mathbf{z}|$ allocated BRS vehicles. The following lemmas reveal some key properties of $Q(\mathbf{z})$ and $f(|\mathbf{z}|)$.

Lemma 4.2. *Let $\mathbf{z}^{(1)} = \{z_i^{(1)} | i \in I^S\}$ and $\mathbf{z}^{(2)} = \{z_i^{(2)} | i \in I^S\}$ be two vectors of first-stage decision variables, which satisfy $z_i^{(1)} \le z_i^{(2)}$ for any $i \in I^S$. Then, $Q(\mathbf{z}^1) \ge Q(\mathbf{z}^2)$.*

Proof. Because $z_i^{(1)} \le z_i^{(2)}$, $z_i^{(1)} \in \{0,1\}$, and $z_i^{(2)} \in \{0,1\}$ for any $i \in I^S$, an immediate observation is that $Q(\mathbf{z}^2)$ is a relaxed version of $Q(\mathbf{z}^1)$, where some values in $\mathbf{z}^{(2)}$ are converted from 0 to 1 in $\mathbf{z}^{(1)}$. We can thus have $Q(\mathbf{z}^1) \ge Q(\mathbf{z}^2)$. □

Lemma 4.2 reveals an important *nonincreasing* property of $Q(\mathbf{z})$ that adding more BRS on an arbitrary solution **z** does not increase the optimal value of $Q(\mathbf{z})$. This property is straightforward to understand since more preallocated trains can certainly transport delayed passengers more quickly. We will show in the next section that this property can derive an improved optimality cuts in Benders decomposition framework.

Lemma 4.3. *The optimal cost $f(|\mathbf{z}|)$ is an also nonincreasing function with respect to $|\mathbf{z}|$.*

Proof. First, according to Lemma 4.1, we know that the values of the first-stage decision variables **z** are unrelated to the feasibility of the second-stage BARP model. Thus, we assume two different decision variables $\bar{\mathbf{z}}$ and $\hat{\mathbf{z}}$ that satisfy Equation (3) and $|\bar{\mathbf{z}}| < |\hat{\mathbf{z}}|$. In addition, we define $\bar{\mathbf{z}}^* = argmin_{\mathbf{z}:|\mathbf{z}|=|\bar{\mathbf{z}}|}\ Q(\mathbf{z})$ and $\hat{\mathbf{z}}^* = argmin_{\mathbf{z}:|\mathbf{z}|=|\hat{\mathbf{z}}|}\ Q(\mathbf{z})$, indicating that $Q(\bar{\mathbf{z}}^*) = f(|\bar{\mathbf{z}}|)$ and $Q(\hat{\mathbf{z}}^*) = f(|\hat{\mathbf{z}}|)$. Here, we notice that the variable vector $\bar{\mathbf{z}}^*$ can be transformed into another variable vector $\bar{\mathbf{z}}'$, where $|\bar{\mathbf{z}}'| = |\hat{\mathbf{z}}|$, by changing the values of some variables z_i from 0 to 1. According to Lemma 4.2, we have the following inequality $Q(\bar{\mathbf{z}}^*) \ge Q(\bar{\mathbf{z}}')$. Meanwhile, we notice that $\hat{\mathbf{z}}^*$ is the optimal solution that yields the minimum objective value given a solution in which $|\hat{\mathbf{z}}|$ values of variables z_i, $i \in I^S$, are equal to 1. Through these analyses, we obtain the following inequality:

$$f(|\bar{\mathbf{z}}|) = Q(\bar{\mathbf{z}}^*) \ge Q(\bar{\mathbf{z}}') \ge Q(\hat{\mathbf{z}}^*) = f(|\hat{\mathbf{z}}|). \quad \square$$

Lemma 4.4. *There exist finite values L and LB that imply the lower bounds for $Q(\mathbf{z})$ and Model (19), respectively.*

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Proof. From the property described in Lemma 4.3, we can immediately conclude that the minimum value L of $Q(\mathbf{z})$ is obtained when $|\mathbf{z}| = |I^S|$, providing a lower bound on $Q(\mathbf{z})$. In other words, $z_i = 1$ for any $i \in I^S$, indicating that all BRS are allocated. Meanwhile, we note that a lower bound for the BARP model can also be given by $LB = \sum_{i \in I^S} \lambda_d c_i * 0 + Q(\check{\mathbf{z}})$, where $\check{\mathbf{z}} = \{z_i = 1, \ \forall i \in I^S\}$. □

5. Solution Approaches

We consider the BRS allocation plan $\mathbf{z}$ as the first-stage decision and treat the train rescheduling variables $\mathbf{x}$ and passenger flow variables $\mathbf{y}$ as the second-stage decisions (which depend on $\mathbf{z}$). Among these decision variables, we wish to find an optimal BRS allocation plan $\mathbf{z}^*$ that minimizes the expected cost over a number of unexpected disruptive situations with respect to uncertain passenger demands. By using a large amount of historical passenger demand data, the two-stage stochastic programming problem can be reformulated as an MILP model via the sample average approximation technique; however, in practice, this leads to enormous numbers of both integer and continuous variables.

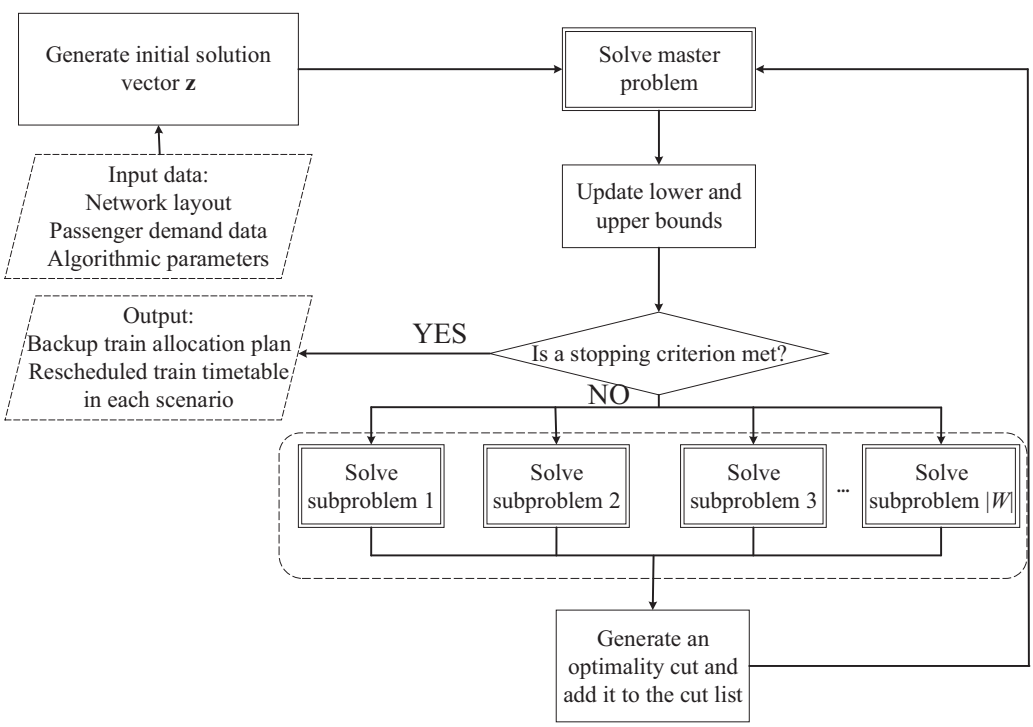
Benders decomposition is an efficient decomposition technique for use in the field of stochastic programming because of the block structure of the model constraints (Birge and Louveaux 2011). However, standard Benders cuts cannot be directly applied in our problem because each subproblem should be continuous in order to make use of linear programming duality theory (Rahmaniani et al. 2017). In addition, standard Benders cuts could be weak in our problem formulation as a result of the binary variables $z_i \in \{0, 1\}, i \in I^S$, in the first stage (Lu et al. 2017). Meanwhile, we note that the first stage of the BARP is a relatively small-scale problem. More important, there always exists a lower bound that can be easily computed, according to Lemma 4.4. This motivates us to decompose the BARP into a master problem (which is easier to solve) and a set of independent subproblems. Thus, the subproblems, one for each stochastic scenario, can be handled simultaneously through parallel computing to enable the full utilization of modern multicore high-performance computing technologies. To further improve the convergence efficiency, we develop a standard integer L-shaped method and an improved L-shaped method with enhanced optimality cuts. In each iteration, a set of optimality cuts is added into the master problem to improve the lower bound until a near-optimal solution can be generated. A general flowchart of the proposed decomposition-based solution approach is presented in Figure 7.

We next present our algorithmic framework in two parts. Section 5.1 introduces the proposed integer L-shaped method and an improved integer L-shaped cut. Section 5.2 describes the speed-up procedures for improved performance of the overall algorithmic framework.

5.1. Improved Integer L-Shaped Cut

Integer L-shaped method is an extension of basic Benders decomposition to a stochastic integer version. For a two-stage stochastic programming problem with binary

Figure 7. General Framework of Our Integer L-Shaped Method



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first-stage variables and integer second-stage variables, the integer L-shaped method has been shown to be an effective approach (Laporte and Louveaux 1993, Carøe and Tind 1998). Furthermore, the integer L-shaped method has been successfully applied to problems in numerous research fields, such as vehicle routing problems (Laporte et al. 2002), network design problem (Riis and Andersen 2002), resource allocation problem (Zhang et al. 2016), and the modeling of electric power systems (Zou 2017), for which the cuts are customized in accordance with the particular model properties.

To implement the integer L-shaped method for our optimization problem, we note that the original model (19) in Section 4 is equivalent to the following model:

$$\min_{\mathbf{z}} \sum_{i\in I^S} \lambda_d c_i z_i + \theta \tag{20}$$

$$\text{s.t.} \sum_{i\in I^S} z_i \le K_b, \tag{21}$$

$$\theta \ge Q(\mathbf{z}), \tag{22}$$

$$z_i \in \{0,1\}, i \in I^S. \tag{23}$$

Here, if we take a binary vector $\mathbf{z}$ as the argument, $Q(\mathbf{z})$ is a real-valued function (or value function), given as follows:

$$\begin{aligned} Q(\mathbf{z}) &= \sum_{w\in W} p^w Q^w(\mathbf{z}) \\ &= \sum_{w\in W} p^w \Big[\min_{\mathbf{x},\mathbf{y}} \{F(\mathbf{y}) : \text{ Constraints(6)}, \\ &\quad \mathbf{x} \in \mathcal{X} \cup \{0,1\}, \mathbf{y} \in \mathcal{Y}\} \Big], \end{aligned} \tag{24}$$

where $Q^w(\mathbf{z})$ is the value function of the subproblem regarding to scenario w (termed as SP_w), given passenger demand in scenario w and vector $\mathbf{z}$ as input; $\mathcal{X}$ and $\mathcal{Y}$ denote the feasible sets of network flow variables $\mathbf{x}$ and $\mathbf{y}$, respectively, defined by Constraints (5)–(7), (11), (14), (17), and (18). The basic idea of the implemented L-shaped method is to relax Constraint (22) and successively add cuts in the $(\mathbf{z},\theta)$ space to better approximate the shape of $Q(\mathbf{z})$ (Angulo et al. 2016). In each iteration, if the solution $(\mathbf{z}^*,\theta^*)$ satisfies $\theta^* = Q(\mathbf{z}^*)$, then an optimal solution has been found; otherwise, feasibility cuts and optimality cuts are added into the master problem in accordance with the computational results for each subproblem. For the problem under consideration, only optimality cuts are needed, because the second-stage BARP models are always feasible given any first-stage decision set $\mathbf{z}$, according to Lemma 4.1 in Section 4. Let L denote a lower bound on $Q(\mathbf{z})$; then, our master problem (MP) is formulated as follows:

$$\text{(MP)} \quad \min_{\mathbf{z}} \sum_{i\in I^S} \lambda_d c_i z_i + \theta \tag{25}$$

$$\text{s.t.} \sum_{i\in I^S} z_i \le K_b, \tag{26}$$

$$\mathcal{L}(\mathbf{z}, Q(\mathbf{z}), L, \theta) \ge 0, \tag{27}$$

$$z_i \in \{0,1\}, i \in I^S. \tag{28}$$

In MP, $\mathcal{L}(\mathbf{z}, Q(\mathbf{z}), L, \theta) \ge 0$ is the set of valid optimality cuts generated in each iteration by solving all subproblems. In the L-shaped method, the optimality cut in Equation (27) actually underestimates $Q(\mathbf{z})$ on the feasible region of $\mathbf{z}$, whereas the estimation is tight at $\mathbf{z}^*$. Let $S(\mathbf{z}^*) = \{i | z_i^* = 1\}$. The standard optimality cuts of the integer L-shaped method in Laporte and Louveaux (1993) is defined as follows:

$$\theta \ge (Q(\mathbf{z}^*) - L)\left(\sum_{i\in S(\mathbf{z}^*)} z_i - \sum_{i\notin S(\mathbf{z}^*)} z_i - |\mathbf{z}^*| + 1 \right) + L, \tag{29}$$

where $\mathbf{z}^*$ is the optimal integer solution obtained from MP, $Q(\mathbf{z}^*)$ is the associated cost for the second-stage optimization problem, and $|\mathbf{z}^*|$ corresponds to the number of variables that are equal to 1. The constraints expressed in (29) are valid inequalities with respect to $Q(\mathbf{z})$, which are tight at $\mathbf{z}^*$. The proof of the latter statement can be found in Laporte and Louveaux (1993). The standard integer L-shaped method for solving a BARP model is summarized in Algorithm 1 (see Online Supplement E.1).

The main issue with respect to the standard integer optimality cut (29) is that it is tight only at the current solution $\mathbf{z}^*$, and it provides no information for all other solutions (i.e., the right-hand side of (29) is equal to or less than the lower bound L). To overcome this issue, we next propose an improved version of the standard integer L-shaped method given the nonincreasing property of $Q(\mathbf{z})$ in Lemma 4.2.

Proposition 5.1. *Given a feasible binary solution $\mathbf{z}^*$ and the corresponding optimal value $Q(\mathbf{z}^*)$, the cut of the form*

$$\theta \ge (Q(\mathbf{z}^*) - L)\left(1 - \sum_{i\notin S(\mathbf{z}^*)} z_i\right) + L \tag{30}$$

is a valid inequality for (MP) and is also tight at $\mathbf{z}^*$.

Proof. See Online Supplement E.2. □

Corollary 5.1. *Inequality (30) can always provide tighter lower bound than the standard L-shaped cut (29).*

This corollary can be easily derived as follows: the right-hand side of (30) still equals $Q(\mathbf{z}^*)$ with $\mathbf{z}^*$ as input; if some values z_i $(i \in I^S)$ in $\mathbf{z}^*$ are converted from 1 to 0, Inequality (30) can return a much better lower bound $Q(\mathbf{z}^*)$ for θ, instead of a value that is less than (or equal to) the global lower bound L in standard L-shaped cut (29); otherwise, we can still guarantee that the right-hand side of (30) is larger than the global lower bound L. The underlying idea between the new optimality cut (30) is that we generate a "flat plane" in

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the space of θ and $\mathbf{z}$ according to the nonincreasing property of $Q(\mathbf{z})$ in Lemma 4.2, whereas the standard optimality cut only cuts a "point."

5.2. Algorithm Acceleration Techniques

The integer L-shaped method usually produces tight optimality cuts, but the convergence can be very slow. The main difficulty in solving the BARP is related to the hardness of the individual scenario problems SP_w, $w \in W$, which contain both integer and continuous variables. In the following subsections, we develop several speed-up techniques to improve the proposed L-shaped method, with the aim of enhancing the computational efficiency.

5.2.1. Improved Optimality Cuts for Adjusting the Search Direction. Preliminary tests of solving the BARP models using the integer L-shaped method reveal that the optimal solution returned for the relaxed MP always varies only slightly around the solution $(\mathbf{z}^*, \theta^*)$ found in the previous iteration. This behavior evidently reduces the computational efficiency of the proposed method and does not yield good improvements with respect to the upper and lower bounds. In the beginning of the algorithm, the integer L-shaped cut does not allow for different priorities between $\mathbf{z}$ variables, causing many ineffective solutions to be found during the iterative process. For example, a series of low-quality solutions $\mathbf{z}$, where $|\mathbf{z}| = 1$, will be generated in the initial iterations, which may take a considerable amount of time and generate only moderate improvements with respect to the upper bound. Therefore, we next investigate how to modify the optimality cuts to improve the computational efficiency of our method.

Proposition 5.2. *Let $\{\xi_i | i \in I^S\} \in \mathbb{R}^{|I^S|}$ be a vector of $|I^S|$ real values satisfying $\xi_i \geq 1$ for each $i \in I^S$, and let $\mathbf{z}^*$ be a feasible binary solution returned by MP. Then, the cut of the form*

$$\theta \geq (Q(\mathbf{z}^*) - L)\left(\sum_{i \in S(\mathbf{z}^*)} \xi_i z_i - \sum_{i \notin S(z^*)} \xi_i z_i - \sum_{i \in I^S} \xi_i z_i^* + 1\right) + L \tag{31}$$

is a valid inequality for (MP) and is tight at $\mathbf{z}^*$.

Proof. See Online Supplement E.2. □

Proposition 5.2 enables us to properly set the values of the coefficients ξ_i. These values essentially prioritize each variable z_i. In this way, we can improve the search direction for a feasible solution by adjusting the values of ξ_i in accordance with the particular characteristics of the studied problem, which is similar to a warm-start strategy to find a good feasible solution. In our practical test case, because there are usually more passengers traveling in the upward direction during morning peak hours, meaning that BRS operating in this direction can serve more stations and carry more passengers, it is usually preferable to allocate BRS to stations in the upward direction rather than in the downward direction. In accordance with this characteristic, we can set the values of the ξ_i such that they are proportional to the passenger demand at each station i. In each iteration, the values of the ξ_i can also be updated in accordance with the objective function value. In other words, the ξ_i values can be learned from the information provided by the solutions found during previous iterations. In addition, Proposition 5.2 illustrates that, regardless of the values of the ξ_i, the algorithm will converge to the theoretical optimal value for the BARP model (as presented in Section 4) if $\xi_i \geq 1$ holds. The reason for this convergence behavior is that the number of decision variables $\mathbf{z}$ in the first stage is finite, and each optimality cut of the form given in (31) rejects a solution $(\mathbf{z}^*, \theta^*)$ if $\theta^* < Q(\mathbf{z}^*)$. In the implementations, we apply both cuts (30) and (31) to reinforce the convergence of the algorithm, as the master problem contains only $|I^S|$ binary variables. In theory, the optimal solution will eventually be found through the iterative process.

5.2.2. Continuous Subgradient Cuts. An optimality cut of the form given in (29)–(31) relies on an exact evaluation of $Q(\mathbf{z})$, which can be very time consuming because $Q(\mathbf{z})$ is essentially a complex mixed-integer linear program. In addition, according to our computational experience, the linear relaxation of $Q(\mathbf{z})$, denoted by $Q_{lp}(\mathbf{z})$, is very tight because of the structure of our mathematical formulation. In some particular cases of passenger demand distributions, the optimal value of $Q(\mathbf{z})$ can even be equal to $Q_{lp}(\mathbf{z})$. This situation motivates us to first compute the linear relaxation of $Q(\mathbf{z})$, which may be very efficient because $Q_{lp}(\mathbf{z})$ is a linear programming problem. Then, we evaluate whether $\theta^* < Q_{lp}(\mathbf{z}^*)$ holds and add continuous subgradient cuts (or Benders cuts, for continuous problems) to MP to remove the candidate solution $(\mathbf{z}, \theta)$. Otherwise, we continue to compute $Q(\mathbf{z}^*)$ in our implementation of the integer L-shaped method until an optimal solution is found (i.e., $\theta^* = Q(\mathbf{z}^*)$) or some other stopping criterion (as specified in the following section) is satisfied.

Under each scenario, the continuous subgradient cuts are generated from the dual of the linear formulation $Q_{lp}(\mathbf{z})$ by following the adopted Benders decomposition scheme in Figure 7. These cuts are essentially the subgradient cuts related to convex optimization. Specifically, we compute $Q_{lp}(\mathbf{z}^*)$ and obtain the optimal values $\hat{x}^w$ and $\hat{y}^w$ as well as the associated dual variables $\pi(w)$ for each subproblem $w \in W$. Because the BARP solution is always feasible and bounded,

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we generate the Benders optimality cuts in the following form:

$$\theta \geq \sum_{w \in W} p^w (\mathbf{b}(w) - \mathbf{c}(w)\mathbf{z})^{\mathrm{T}} \pi(w), \tag{32}$$

where $\mathbf{b}$ and $\mathbf{c}$ are the associated costs with respect to the constraints with form $y(w) + \mathbf{c}(w)\mathbf{z} \geq \mathbf{b}(w)$ in each primal subproblem w.

In essence, the key idea of the continuous cuts presented in the previous paragraph is to use $Q_{lp}(\mathbf{z})$ as a proxy for $Q(\mathbf{z})$ and check the feasibility of a candidate solution $\mathbf{z}^*$ by comparing $Q_{lp}(\mathbf{z}^*)$ with θ^*, thereby preventing unnecessary computations of $Q(\mathbf{z})$, which are more time consuming. The proposed treatment is similar to the sequential approximation schemes presented in Sen and Higle (2005), Angulo et al. (2016), and Gade et al. (2014), in which the second-stage cost function is approximated by linear programs and iteratively strengthened with additional cuts. In our problem, because the linear relaxation of $Q(\mathbf{z})$ is quite tight, this approach is very useful in acquiring a better-quality solution much more efficiently than directly solving the corresponding MILP models in each iteration.

In the improved L-shaped method (ILM) based on our Benders decomposition approach, we apply both the promising optimality cuts (30) and (31) and the continuous cuts (32) generated from each subproblem to the MP in each iteration until an optimal (or near-optimal) solution is found or (at least) one of the stopping criteria is reached. We present the implementation details of our ILM algorithm in Algorithm 2 (see Online Supplement E.3).

6. Computational Experiments

In this section, we assess the computational performance of the various models and solution approaches presented in this paper. Section 6.1 compares the performance of the proposed solution methodologies (CPLEX, standard integer L-shaped method, and our ILM) for solving the BARP models. Section 6.2 investigates the benefits of stochastic optimization for the BARP. Section 6.3 analyzes the trade-off between investment cost and QoS improvement. The purposes of these experiments are threefold. First, we identify the most effective approach for solving the BARP models. Second, we compare the outcomes of this best approach with the practical solutions (with no BRS allocation before 2019) and heuristically generated solutions to show the potential improvement that can be achieved when applying the proposed methodology in practice. Third, we report quantitative computational results to support rail managers when making decisions on the BRS allocation. The experiments were conducted on a high-performance computer system with a total of 32 Intel Xeon 6140 processors and 48 GB of RAM. We also used IBM CPLEX 12.3 Academic Version on the same platform for fair comparisons when solving the subproblems for our Benders decomposition approach.

We tested our models and solution approaches using data collected from the Batong line in the Beijing metro network. The Batong line is a bidirectional commuter rail line (with two parallel unique-directional tracks) with a total length of 18.94 km and 13 physical stations connecting the center of Beijing and Tongzhou District, as shown in Figure 8. For convenience, we use numbers 1–13 to denote the stations in the upward direction and numbers 14–26 to denote the stations in the downward direction. According to the designed layout of the rail tracks (see Online Supplement E.4), the Batong line has two depots, one located near station 1 and another near station 14. In addition, most stations of Batong line are above ground and are designed with sidetracks, which enable train dispatchers to reserve BRS at most stations (except for stations 6, 18, and 21); in other words, the set of stations with train storage lines is $I^S = I \backslash \{6, 18, 21\}$.

We collected and processed data on accidents that occurred during daily operations from January 2017 to July 2017. During this time period, a total of 65 incidents (involving equipment failures, oversaturated passenger flows and a suspected fire, etc.) that disrupted the original timetable were recorded (in these railway data, the longest incident was caused by a shutdown of Automatic Train Protection equipment on a train, which lasted approximately 35 minutes). We generate a set of test instances with different numbers of scenarios (the detailed procedure appears in Online Supplement E.5). For each scenario, the input data for the second-stage model include the positions of the existing trains, the static passenger demand, and the time-dependent passenger demand (the open-source data and codes are available in the IJOC GitHub repository, available at https://dx.doi.org/10.5281/zenodo.6892548). An illustration of the time-dependent passenger demand data, extracted from one specific scenario, is shown in Figure 9. This illustration presents the number of arriving passengers in each time unit (1 minute) from 8:00 a.m. to 10:00 a.m. on a weekday at each station of the Batong line.

6.1. Comparison Among Various Algorithms on BARP

In this section, we compare the performance of our decomposition based algorithms with the benchmark CPLEX solver in solving the BARP-S and BARP-D models. We recall that the BARP-S model does not consider the arrival of additional passengers during the recovery phase, whereas the BARP-D model considers both static and dynamic passenger demands. Along the chosen bidirectional transit line with $|I| = 26$ stations, instances were generated with a 2-hour time horizon. The

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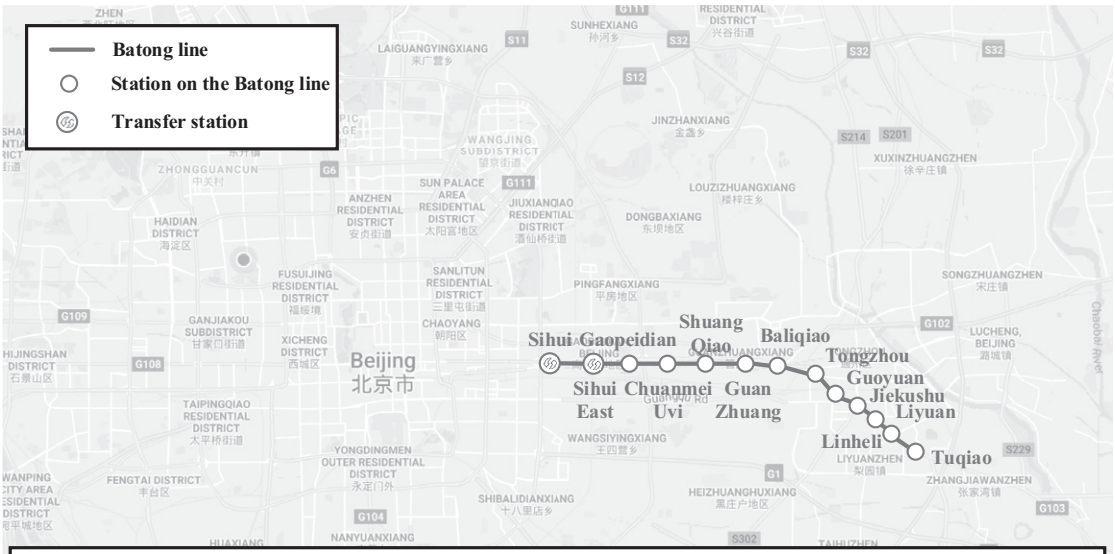
Figure 8. Illustration of the Batong Line in Beijing



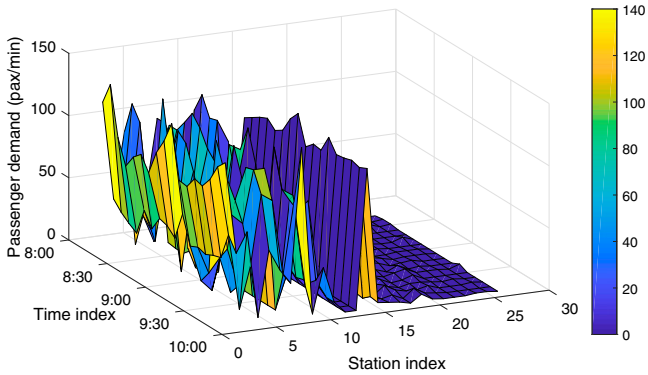
Indexes of stations	Tuqiao	Linheli	Liyuan	Jiekushu	Guoyuan	Tongzhou	Baliqiao	Guan Zhuang	Shuang Qiao	Chuanmei Uvi	Gaopeidian	Sihui East	Sihui
Upward direction	1	2	3	4	5	6	7	8	9	10	11	12	13
Downward direction	26	25	24	23	22	21	20	19	18	17	16	15	14

minimum time unit was set to 10 seconds (as the dwell times of trains in Beijing metro are adjustable in 10-second increments), and thus, a total of 720 timestamps were considered in the experiments. The investment cost for each BRS was set to 1.0×10^6 to test the balance between the investment cost and the QoS for passengers in our objective function.

6.1.1. Experiments on the BARP-S Model. We compare the computational performance of three methods for solving the BARP-S model: a direct solution method using the default CPLEX settings (with the parallel computing option turned on), the standard integer L-shaped method, and our improved L-shaped method presented in Section 5. We denote them by CPLEX-D, LSM, and ILM, respectively. Here, it is worth mentioning that the Benders decomposition technique embedded in CPLEX cannot be used to solve the BARP models because it requires a continuous recourse function (i.e., no integer variables exist in the second-stage model). Thus, we consider only the default mixed-integer programming solver of CPLEX as a benchmark. The stopping criteria were set as follows:

1. The instance is solved to within a given optimality gap tolerance (set to 1% in our experiments). The percentage optimality gap is calculated as $100\% \times (UB$

Figure 9. (Color online) Passenger Demand Data Collected on the Batong Line from 8:00 a.m. to 10:00 a.m. (from One Data Sample)



Note. Pax, passenger.

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Table 2. Performance Comparison of the Various Approaches Proposed to Solve BARP-S

Instance index	Solution method	Fleet size of BRS	ATT (s)	Objective value	CPU time (s)	Gap (%)
S-1	CPLEX-D	10	2,305.9	1.37×10^8	1,154.2	0.23
	LSM	7	2,395.6	1.39×10^8	2,031.4	0.82
	ILM	8	2,359.3	1.38×10^8	1,217.2	0.78
S-3	CPLEX-D	11	2,434.3	1.43×10^8	3,055.08	0.02
	LSM	10	2,489.6	1.45×10^8	4,874.08	0.85
	ILM	13	2,397.4	1.43×10^8	2,812.00	0.83
S-5	CPLEX-D	13	2,409.2	1.40×10^8	838.05	0.01
	LSM	13	2,447.2	1.42×10^8	7,200.08	1.42
	ILM	13	2,409.2	1.40×10^8	4,055.08	0.98
S-10	CPLEX-D	13	2,410.4	1.40×10^8	1,569.74	0.04
	LSM	13	2,429.4	1.41×10^8	3,055.08	1.33
	ILM	13	2,410.4	1.40×10^8	2,012.08	0.85
S-20	CPLEX-D	1	2,863.9	1.52×10^8	7,242.0	NAN
	LSM	13	2,465.7	1.43×10^8	7,200.08	6.32
	ILM	13	2,465.7	1.43×10^8	3,055.08	0.85
S-25	CPLEX-D	—	—	—	—	—
	LSM	6	2,664.7	1.46×10^8	7,200.08	8.85
	ILM	13	2,474.3	1.43×10^8	7,200.08	1.88
S-30	CPLEX-D	—	—	—	—	—
	LSM	1	2,698.7	1.43×10^8	7,159.41	12.58
	ILM	8	2,489.6	1.39×10^8	7,080.5	1.73%

Notes. In this paper, the term "NAN" indicates that CPLEX returns no optimality gap within the computational time limit. The term "—" indicates that no feasible solution is obtained within the computational time limit.

$-LB)/UB$, where UB denotes the best upper bound and LB the best lower bound obtained up to the current iteration.

2. The total time required to solve the current instance exceeds a given computational time limit, set to two hours in this study.

Table 2 presents the results obtained for various sets of stochastic scenarios. The columns of Table 2 present the instance index S-$|W|$ (representing the passenger type—that is, static passenger demand (S) —and the number of stochastic scenarios $|W|$), the solution method, the fleet size of allocated BRS used in the returned solution, the objective function value, the average travel time (ATT) of the involved passengers (in seconds), the CPU time (in seconds), and the optimality gap (in percent). Using these parameter settings for the studied instances, we present a series of experimental results with an increasing number of stochastic scenarios. Here, to quantify the passenger ATT (which takes different values across different scenarios), we present the expected values in Table 2.

From the experimental results in Table 2, we conclude that CPLEX is an efficient solver only for small-scale instances (e.g., S-1, with only one scenario, and S-3, with three scenarios) and returns high-quality solutions for these instances, whereas its performance becomes unacceptable as the number of scenarios continues to increase. For instance S-20, only one feasible solution with no optimality gap is returned by CPLEX after a two-hour computation time. It is evident that the passenger ATT achieved with the feasible solution returned by CPLEX is too long, because only one backup vehicle is allocated. The computation time of CPLEX also reveals a similar trend (i.e., good performance for small-scale instances only). CPLEX cannot return any feasible solution after a two-hour computation time for instance S-25 (denoted by the em dash in Table 2). A possible reason is that CPLEX uses a cutting plane method combined with a heuristic search to solve the proposed model. CPLEX cannot handle such a large-scale problem with integer variables by adding valid inequalities without effective ad hoc decomposition techniques.

Regarding the Benders decomposition–based algorithms (LSM and ILM), their performances are much more stable than those of CPLEX. For most instances, a near-optimal solution is computed with an optimality gap of less than 2% and within two hours of computation time using both LSM and ILM. With the help of our algorithmic speed-up techniques and the implementation of parallel computing for the scenario-based subproblems, LSM and ILM can obtain high-quality solutions for instances S-25 and S-30 (in which CPLEX cannot return even a feasible solution), whereas ILM obtains much better performance, as the passenger ATT of ILM is evidently decreased compared with LSM. Furthermore, we observe that ILM requires a much shorter computation time when the number of scenarios is fewer than (or equal to) 20, and ILM produces better results than LSM for nearly all tested instances. The returned optimality gap of ILM is always less than 2%, thus certifying the effectiveness of our decomposition

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scheme with improved optimality cuts when dealing with a large number of scenarios.

6.1.2. Experiments on the BARP-D Model. We next report computational experiments based on the BARP-D model under dynamic passenger demands. The scenario-based passenger demands were extracted from practical passenger OD data for the Batong line. Aside from the passenger demand data, all other settings were the same as those in the experiments presented in Section 6.1.1. In the following, we compare the results of CPLEX-D, LSM, and ILM.

We present the computational results of a total of 30 instances in Table 3 and Tables 9 and 10 (the latter two tables appear in Online Supplement E.6), obtained for various numbers of stochastic scenarios when using the BARP-D model. The information reported in each column is the same as in Table 2. Overall, we observe that the performance of CPLEX in solving the BARP-D model is much worse than that for the BARP-S model (as shown in Table 2). CPLEX uses the maximum allowed computation time to solve the BARP-D model for all tested instances and cannot return a good-quality solution for most instances: for instances D-3 and D-10, solutions with no BRS and a very large ATT are generated, and for instances D-20, D-25, and D-30, no feasible solution can be obtained by CPLEX. The reason for this poor performance is straightforwardly related to the complexity of BARP-D, which is much higher than that of BARP-S as a result of the consideration of time-dependent passenger demands. By comparison, LSM shows relatively good performance for solving small-scale and medium-scale instances. Feasible solutions are always obtained within a two-hour computation time using LSM. However, for large-scale instances (instances with more than 25 scenarios), the performance of LSM is not satisfactory. For instance D-25, the passenger ATT obtained with LSM is much higher than that obtained with ILM; for instance D-30, no feasible solution can be obtained with LSM. A possible reason is that LSM seeks solutions via a conservative approach (i.e., seeking all neighbors from $|\mathbf{z}| = 0$ to $|\mathbf{z}| = 1$, $|\mathbf{z}| = 2, \ldots, |\mathbf{z}| = |I^S|$), leading to slow convergence. We also present the average performance comparison on these 30 instances in Table 3. The most important result is the following: CPLEX can obtain feasible solutions

Table 3. Performance Comparison of Various Approaches Proposed to Solve BARP-D

Instance index	Solution method	Fleet size of BRS	ATT (s)	Objective value	CPU time (s)	Gap (%)
D-1	CPLEX-D	5	3,215.8	2.31×10^8	7,200.3	0.57
	LSM	7	3,209.1	2.33×10^8	2,031.4	1.82
	ILM	5	3,211.4	2.31×10^8	1,217.2	0.01
D-3	CPLEX-D	0	3,569.3	2.53×10^8	7,209.7	10.17
	LSM	5	3,329.4	2.41×10^8	7,209.7	4.82
	ILM	6	3,273.0	2.38×10^8	7,200.4	0.78
D-5	CPLEX-D	9	3,153.8	2.35×10^8	7,214.2	1.55
	LSM	7	3,237.5	2.39×10^8	7,200.3	0.82
	ILM	5	3,209.6	2.35×10^8	7,200.2	0.55
D-10	CPLEX-D	0	3,528.9	2.49×10^8	7,201.6	NAN
	LSM	7	3,273.8	2.38×10^8	7,200.3	8.82
	ILM	8	3,160.4	2.31×10^8	7,200.2	1.78
D-20	CPLEX-D	—	—	—	—	—
	LSM	2	3,316.8	2.38×10^8	7,200.3	3.42
	ILM	8	3,091.9	2.28×10^8	7,200.2	0.78
D-25	CPLEX-D	—	—	—	—	—
	LSM	1	3,402.0	2.43×10^8	7,200.3	5.44
	ILM	9	3,120.8	2.31×10^8	7,200.2	1.26
D-30	CPLEX-D	—	—	—	—	—
	LSM	—	—	—	—	—
	ILM	8	3,120.9	2.27×10^8	7,200.1	2.86
Average	CPLEX-D	12[a]	—	3.31%[b]	7,200.4	—[c]
	LSM	29[a]	—	2.66%[b]	6,858.8	5.37%[c]
	ILM	30[a]	—	—	6,785.4	1.49%[c]

Notes. In this paper, the term "NAN" indicates that CPLEX returns no optimality gap within the computational time limit. The term "—" indicates that no feasible solution is obtained within the computational time limit.

[a]The values represent the number of instances on which the method obtains a feasible solution among all the 30 instances.

[b]The values represent the improvement (in percentage) of ILM with respect to LSM or CPLEX-D, regarding the best objective for instances with feasible solutions.

[c]The values represent the average optimality gaps for instances where feasible solutions are generated (the average optimality gap of CPLEX-D is not presented, as it cannot return the optimality gaps after instance D-10).

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for only 12 instances, whereas our decomposition approaches generate good-quality solutions in nearly all the instances. ILM outperforms CPLEX and LSM for all tested instances and always computes near-optimal solutions. The passenger ATT obtained with ILM is markedly reduced compared with the solutions obtained by using the other two approaches. The optimality gaps of the solutions computed by ILM for all tested instances are always below 3%.

6.2. Value of Stochastic Optimization

6.2.1. Experiments with Different BRS Allocation Strategies. We now compare the performance of four strategies for computing BRS allocation plans. The first strategy involves no BRS in the system, and the second strategy is to randomly generate a set of stations at which we allocate BRS. The third strategy is to use our optimization-based BARP to allocate the BRS. Using the same fleet size of BRS with the third strategy, we present an additional strategy, termed a *partial allocation*, to allocate these BRS uniformly along the line. In our experiments, we first generated a set of BRS allocations using each strategy, and we then used each BRS allocation plan $\mathbf{z}$ as input to compute the optimal value of the recourse function $Q(\mathbf{z})$ as well as the passenger ATT.

Table 4 presents the fleet size of allocated BRS, the passenger ATT, and the value of our objective function. The test instances are the same as those in Section 6.1.2. We see that the passenger travel time with no BRS is the worst. This is expected because of the large difference between the oversaturated passenger demand and the limited train capacity. With the randomly generated BRS allocation plan, the passenger travel time is evidently reduced (by approximately 7%, on average) compared with the strategy with no BRS; however, the random allocation strategy requires a large fleet of BRS, resulting in the largest objective function value for four of the five tested instances. With the same scales of allocated BRS, our optimization-based approach evidently outperforms the partial allocation strategy in all the test instances, highlighting the importance of the optimized BRS allocation plan for reducing the ATT of delayed passengers. In summary, we conclude that the overall performance can be noticeably improved with our optimization-based approach.

6.2.2. Experiments with Different Numbers of Stochastic Scenarios. The experiments reported in Section 6.1.2 show that our customized ILM can be used to solve 30-scenario instances with the BARP-D model, whereas CPLEX can handle only up to 10 scenarios (see Table 9 in the online supplement). We next investigate the following question: How can we choose the best number of scenarios to balance the computational intensity and optimization performance? To answer this question, we conducted the following set of experiments. Using the solutions $\mathbf{z}$ of ILM from the instances in Table 3 as input, we computed the least expected passenger travel time $Q(\mathbf{z})$ as well as the objective function values of BARP-D across the overall 65 incidents. Table 5 compares the performance of seven solutions (from 5 scenarios up to 35 scenarios) to a baseline instance with no BRS. The results yield two observations. First, applying our ILM algorithm to 30 scenarios (i.e., D-30) results in a noticeably better solution than the baseline instance with no BRS and the instance with only 5 scenarios (i.e., D-5), with a reduction of 6.58% and 4.22%, respectively, in the objective value. Compared with the baseline with no BRS, the passenger ATT using the BRS plan for D-30 can be remarkably reduced by 8.63%, from 3,414.5 to 3,120.9. The results reveal the benefits of stochastic programming for the improved QoS to passengers. Second, when the number of scenarios increases from 5 to 20, the objective function value of ILM sharply decreases, whereas the objective function value computed with ILM remains almost unchanged when the number of scenarios is above 20. This observation indicates that at least 20 scenarios are required to capture the variations

Table 4. Comparison Among the Following Strategies: No BRS, Random Allocation of BRS, Partial Allocation of BRS, and Optimization-Based BRS Allocation

Performance	Method	D-5	D-10	D-20	D-25	D-30
Fleet size of BRS	No BRS	0	0	0	0	0
	Random allocation	13	11	11	14	13
	Partial allocation	5	8	8	9	8
	BARP-based approach	5	8	8	9	8
ATT (s)	No BRS	3,400.0	3,396.1	3,401.1	3,412.8	3,409.4
	Random allocation	3,226.0	3,212.1	3,274.6	3,281.0	3,266.8
	Partial allocation	3,260.1	3,283.4	3,302.7	3,314.4	3,295.3
	BARP-based approach	**3,209.6**	**3,160.4**	**3,091.9**	**3,120.8**	**3,120.9**
Objective value ($\times 10^8$)	No BRS	2.44	2.41	2.42	2.43	2.39
	Random allocation	2.44	2.38	2.44	2.47	2.42
	Partial allocation	2.39	2.41	2.43	2.45	2.39
	BARP-based approach	**2.35**	**2.31**	**2.28**	**2.31**	**2.27**

Note. The bold value for each instance represents the minimum value.

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Table 5. Performance Comparison of BRS Allocation Plans Under Different Numbers of Scenarios

Instance	Baseline	D-5	D-10	D-15	D-20	D-25	D-30	D-35
ATT (s)	3,414.5	3,279.2	3,287.9	3,195.8	3,123.7	3,123.7	3,120.9	3,120.9
Objective value ($\times 10^8$)	2.43	2.37	2.37	2.32	2.28	2.28	2.27	2.27
Objective improvement (%)	0.00	2.50	2.50	4.53	6.17	6.17	6.58	6.58

of the passenger demands when solving the BARP-D model.

6.3. Sensitivity to Key Parameters

The primary factor affecting the QoS for passengers in an urban rail transit system is their travel time (Wang 2014), especially in the recovery phase of a disruptive event. Therefore, there is an emerging requirement for rail managers to quantitatively analyze the relationship between the BRS investment cost and the QoS offered to passengers. In the following subsections, we analyze the trade-off between these two objectives within a multiobjective optimization (MOOP) context. Specifically, we next conduct two sets of experiments, one based on an ϵ-constraint method and the other on a weight sum method—two widely used methods in MOOP (Ibarra-Rojas et al. 2014, Cooper et al. 2020).

6.3.1. Analyzing the Biobjective Function via an ϵ-Constraint Method. In the planning of BRS, the information about the trade-off between the QoS and the investment cost is valuable for helping railway managers to make better decisions. Our approach can provide decision makers with an explicit approximation of the preference function or satisfaction levels for different criteria (i.e., QoS versus cost). To this end, we employed an ϵ-constraint method, where the investment cost of BRS was imposed as a constraint to the second-stage of BARP model (i.e., $\sum_{i \in I^S} c_i z_i \leq \epsilon$). Then, we used ILM to solve the single-objective BARP-D model for the optimization of QoS. Note that although we assume that the investment cost c_i for allocating each BRS vehicle is the same, the value of ϵ actually represents the fleet size of allocated BRS.

Figure 10 presents the optimal Pareto frontier with the ϵ-constraint method. For decision makers, this kind of graphical solutions can be very illustrative to identify the improvement of QoS with the increased investment cost of BRS. From the results in Figure 10, we can observe that the passenger travel time sharply decreases by introducing additional rolling stocks as backup resources. However, once the fleet size of allocated BRS has increased above 7, the passenger travel time remains relatively steady, which reveals the fact that excess BRS may not benefit to the QoS of passengers because of the limited line capacity from headway constraints. Another interesting observation from Figure 10 is that the Pareto frontier is not necessarily convex—for example, when the fleet sizes of rolling stocks are 5 or 6. This somehow illustrates the advantages of using the ϵ-constraint method, which is able to identify a number of noninferior solutions on a nonconvex boundary. The rail managers can make decisions according to their investment budgets or preferences as well as the satisfaction QoS levels of traveling passengers. The detailed results are listed in Online Supplement E.7.

6.3.2. Analyzing the Biobjective Function via a Weight Sum Method. In this section, we further analyze the trade-off between the investment cost and the QoS by using a weight sum method. In the experiments, we normalized the two objective function components (i.e., investment cost and QoS) and introduced two weight coefficients, $w_c \in [0,1]$ and $w_s \in [0,1]$, to represent the importance of the investment cost and QoS (see Online Supplement E.8 for more details). We used the ILM to optimize the BARP-D model, with the same settings as in Section 6.3.1. To generate the Pareto frontier of BARP-D, we changed w_s from 0 to 1 (and correspondingly changed w_c from 1 to 0) with the increment of 0.1 to obtain the nondominated solutions.

The Pareto frontier of the BARP by the weight sum method is shown in Figure 11. The weight ratio (w_c, w_s) corresponding to each point is marked in Figure 11. From the Pareto frontier by the weight sum method, we find that the optimal values corresponding to different weight ratios may overlap in some cases, such as the points corresponding to weight ratios (0.6,0.4), (0.5, 0.5), (0.4, 0.6). (0.3, 0.7), and (0.2, 0.8). This phenomenon

Figure 10. Pareto Frontier with the ϵ-Constraint Method

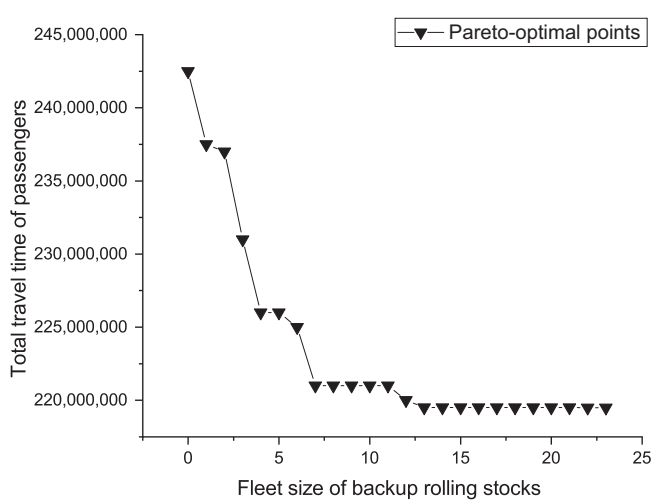

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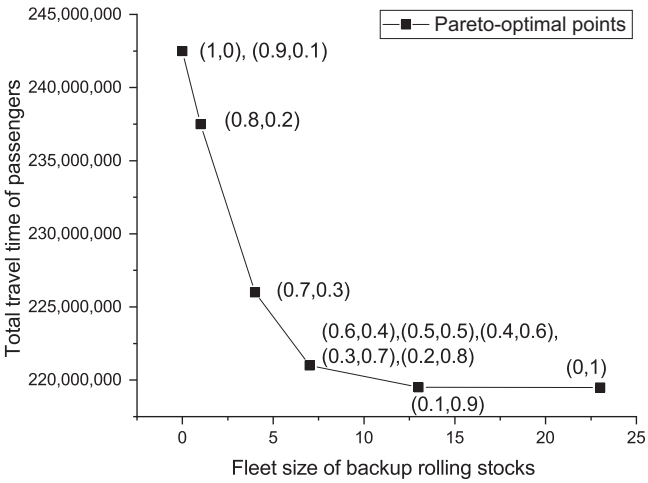
Figure 11. Pareto Frontier with the Weight Sum Method



indicates that a set of weight ratios may produce a single nondominated solution. In addition, as the Pareto frontier by the weight sum method has a convex curve, we observe from Figure 11 that some solutions—for example, BRS fleet size of 5—would have never been found in the obtained Pareto frontier. Thus, the rail managers should also refer to the solutions by the ϵ-constraint method to make the decision of rolling stock allocation plan.

Finally, we visualized the Pareto-optimal solutions that correspond to the fleet sizes of 3 ($w_c = 0.7$, $w_s = 0.3$), 7 ($w_c = 0.6$, $w_s = 0.4$), and 13 ($w_c = 0.1$, $w_s = 0.9$) (see Online Supplement E.9). We can clearly see that the stations in the upward direction are busier than those in the downward direction, and thus, it is preferable to allocate BRS near stations in the upward direction to effectively serve the traveling passengers. From a practical point of view, our research results clearly indicate that BRS should be preferentially allocated to upstream stations in the upward direction (i.e., stations 1–10, as suggested by our results in Table 12 in the online supplement), because these stations usually see large volumes of passengers in the morning peak hours. Since May 2019, the train dispatchers on the Batong line have been allocating BRS to station 5 (Guoyuan Station) and station 8 (Guanzhuang Station) to support recovery from disruptive events during daily operations, which is in accordance with our optimized solutions. With the BRS allocated to these two stations, combined with other train rescheduling strategies (e.g., stop-skipping and reducing dwelling time), the Beijing metro has demonstrated that the minimum headway time between trains is reduced from 2 minutes and 50 seconds to 1 minute and 58 seconds in overcrowded sections in cases of unexpected incidents, which can reduce the congestion of services by approximately 30% and thus mitigate the risk of the novel coronavirus spreading among passengers (Jianfei and Yuan 2020).

7. Conclusions and Future Research

In this paper, we developed two-stage stochastic integer programming models and an improved L-shaped method for jointly optimizing BRS allocation and train rescheduling in the tactical planning phase in response to a disruptive incident. The aim of our work is to enhance the ability of an urban metro line to efficiently recover from unexpected events. Because of the large number of variables required to describe real-world instances, we proposed and compared three solution methods: the default CPLEX solver, a standard integer L-shaped method, and an improved L-shaped method with customized optimality cuts (ILM). The proposed solution algorithms were used to compute near-optimal solutions for a set of instances generated from the real-world data from the Beijing metro network. The computational results indicate that our decomposition methods significantly outperform CPLEX when dealing with large-scale instances. Furthermore, the implementation of the improved optimality cuts in ILM evidently enhances the computational efficiency compared with the standard integer L-shaped method. The latter results highlight the importance of developing customized optimality cuts in Benders decomposition.

Our computational results demonstrate the practical benefits of allocating BRS for recovery from disruptive events that may affect a large number of passengers. By allocating a number of BRS to the investigated metro line such that a good trade-off is achieved between the investment cost and QoS, and using these BRS efficiently in the rescheduled timetable, the average passenger travel time in the incident recovery phase can be significantly reduced (by about 8.63%) compared with a strategy with no BRS. Meanwhile, our computational experiments indicate that the fleet size of allocated BRS and the reduction in passenger travel time do not exhibit a linear relationship. Because of the train headway constraints, the impact of allocating BRS may be minimal when the traffic density is high. On the other hand, the passenger demand pattern also exerts a noticeable impact on the allocation of BRS. From our computational study, we observe that the improvement of QoS to passengers is marginal when the fleet size of BRS exceeds 7. In summary, we conclude that *more may not be better*. Thus, understanding passenger travel patterns is of great importance for metro managers and dispatchers, especially if they wish to improve the QoS in cases of disruptive events during which, for example, the passenger demand is much higher than expected.

This work can be extended in several interesting directions. First, this research addressed the metro rescheduling problem in the recovery phase, once the infrastructure capacity is completely recovered. However, during a disruptive incident, the infrastructure

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capacity of the affected line is quite limited, and short-turning or other control strategies can be very effective. Thus, further research should be dedicated to extending the two-stage model proposed here to a more general multistage model that includes an uncertain disruption duration and additional control strategies. Second, because of the limited data resources, only 65 real-world incidents from half a year were involved in this study. Our future research will address the BARP over a longer time horizon, with the consideration of passenger demand predictions in the planning phase. Third, the primary aim of this study was to reduce the crowdedness of stations while considering the average travel time for passengers as a measure of the QoS. However, passenger preferences may be varied for different OD pairs, and thus, an interesting idea would be to incorporate a more detailed representation of the distribution and deviations of passenger delay times into the QoS indicator. Finally, this study mainly focused on the rescheduling process for a single transit line, and no in-depth analysis of network dependence, large-scale traffic flows, or special passenger behaviors was performed. More efficient methodologies in the field of stochastic optimization (e.g., simulation techniques or reinforcement learning with state aggregation) will be required to address these challenging issues regarding complex railway networks in the near future.

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Appendix

Table A.1 introduces the basic notation and decision variables used in our mathematical model.

Table A.1. Symbols and Notation

Notation	Description
	The first-stage problem
Parameter	
I	Set of stations along the metro line
$I^S \subseteq I$	Set of stations with storage lines or side/pocket tracks
$\mathbb{T} = \{0, 1, \ldots, T\}$	Set of discrete timestamps
$\mathbb{T}_t = \{0, 1, \ldots, t\}$	Set of discrete timestamps for each $t \in \mathbb{T}$ used for formulating cumulative passenger flows
$t_{\min}$	Minimum train following time headway
$\mathbb{T}_t^{\min} = \{\tau \in \mathbb{T} \mid t \le \tau \le t + t_{\min}\}$	Set of discrete timestamps for each $t \in \mathbb{T}$ used for formulating train headway constraints
A^W	Set of spatial-temporal waiting arcs in the network
A^T	Set of spatial-temporal travel arcs in the network
A^I	Set of spatial-temporal assignment arcs in the network
N	Set of spatial-temporal nodes in the network
n_O	Dummy origin node in the network
$\delta^+(n_{it})$ or $\delta^-(n_{it})$	Set of arcs for which n_{it} is the origin or destination node, respectively
$\sigma^+(n_{it}), \sigma^-(n_{it})$	Set of travel arcs for which n_{it} is the origin or destination node, respectively
i, j	Indexes of stations, $i, j \in I$
c_i	Cost of allocating each backup vehicle to station i
K_b	Fleet size budget for BRS
λ_d	Weight coefficient
Decision variable	
z_i	Binary decision variable representing the number of trains allocated to station $i \in I^S$
	The second-stage problem
Parameter	
$I^E \subseteq I$	Set of stations with dwelling trains at time $t = 0$
W	Set of stochastic scenarios
w	Index of random scenarios, $w \in W$
p^w	Probability of each stochastic scenario w, $w \in W$
t_{ij}	Train travel time between stations i and j
C	Train loading capacity
d_{ij}^w	Static passenger demand with origin station i and destination j in scenario w
d_{ijt}^w	Dynamic passenger demand at time t with origin station i and destination j in scenario w
L_{it}^w	Number of passengers with destination i who plan to alight before time t in scenario w
$Q(\mathbf{z})$	Expected objective function value given first stage decision $\mathbf{z}$
Decision variable	
x_a^w	Equal to 1 if arc a is employed by a train in the rescheduled timetable and 0 otherwise
y_a^w	Number of passengers traveling on arc a
α_{it}^w	Number of passengers alighting at node n_{it}

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